

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2024

FINAL REPORT



Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

Prepared by:

David Robichaud¹ and Angela Addison²

¹ LGL Limited environmental research associates
9768 Second Street, Sidney, BC, V8L 3Y8

² North Coast - Skeena First Nations Stewardship Society
309 2 Ave W #363, Prince Rupert, BC, V8J 3T1



25 February 2025

EXECUTIVE SUMMARY

Due to the significant potential impact of the recreational fishery in the North Coast (Areas 3 and 4) on various transboundary salmon stocks, this fishery has a high priority for monitoring. Gathering effort and harvest estimates, along with biological samples and genetic information, is crucial to inform effectively managed fisheries and to monitor indicator stocks. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has operated this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consists of access point angler interviews and aerial overflights, previously for the months of June through August but also including May since 2019. Lodge-based recreational harvest and effort are monitored by DFO via a voluntary logbook program. Survey design and implementation, and analytical methods are similar to and consistent with those used in Area 3 and 4 creel survey monitoring since 2008.

From 1 May to 31 August 2024, a total of 2,808 vessel interviews (involving 9,041 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 247 survey shifts. 30 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon. When possible, length measurements were collected from Pacific Halibut (*Hippoglossus stenolepis*). Survey effort was adequate, as relatively good precision was achieved for the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 13,729 vessel trips. In total, an estimated 19,844 Chinook Salmon were caught, with 9,239 fish kept and 10,605 released. It was estimated that 32,264 Coho Salmon were caught, with 29,253 fish kept and 3,011 released. Total Pacific Halibut catch was estimated to be 25,321 individuals, with 14,957 fish kept (biomass estimated at 95.2 net tonnes) and 10,364 released. Total harvest estimates included 9,510 Lingcod (*Ophiodon elongates*), 53 Yelloweye Rockfish (*Sebastes ruberrimus*), 7,853 other rockfishes (*Sebastes* spp.), and 2,747 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 483 Chinook Salmon (157 in May), collecting length measurements (median 69 cm), adipose fin-clip incidence rates (18.6% were clipped), and scale samples. For Coho Salmon, 725 fish were biosampled (median fork length 62 cm; 0% adipose clipped). Heads were collected from 47 Chinook Salmon and five Coho Salmon for submission to the Salmon Head Recovery Program. Biosampled Pacific Halibut (n=391) had a median length of 80 cm, and average net weights of 7.6 kg and 5.6 kg in Areas 3 and 4, respectively.

During the month of May, fishing effort and Chinook Salmon harvest were estimated at 1,851 boat trips (13% of the May-August effort), and 2,594 fish (28% of the May-August harvest). The Department of Fisheries and Oceans Canada (DFO) is supportive of a May creel survey because genetic samples would help them resolve stock composition and determine exploitation rates on early-run-timing stocks. During the month of May 2024, we biosampled 157 Chinook Salmon, taking 154 scale samples, and processed 12 tagged heads.

Recreational fishing effort in Areas 3 & 4 have been increasing over the last 30 years, as has harvest of Coho Salmon and Lingcod. Lingcod harvest in the last four years are the four highest observed for the available time series, despite a coast-wide reduction in the Lingcod bag limit that started in 2018. By contrast, Chinook Salmon harvests are not showing an increased trend, and may be declining, though Chinook Salmon releases are showing an increasing trend. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. While the increasing catches of some species highlight the need for continued monitoring of the recreational fishery into the future, there is evidence that management measures can keep harvest under control.

TABLE OF CONTENTS

LIST OF TABLES	IV
LIST OF FIGURES	V
INTRODUCTION	1
SCOPE.....	1
GOAL AND OBJECTIVES.....	2
METHODS	3
Angler Activity Patterns.....	5
Catch Per Unit Effort Estimation.....	7
Angler Effort Estimation	7
Lodge Effort Census.....	7
Non-lodge Effort Surveys.....	9
Catch and Harvest Estimation.....	10
Biosampling.....	10
Biomass of Harvested Pacific Halibut	11
Statistical Analyses	11
Trailer Census.....	11
RESULTS.....	12
Angler Interviews.....	12
Angler Activity Patterns.....	12
Catch Per Unit Effort Estimates.....	17
Angler Effort Estimates.....	22
Catch Estimates.....	28
Biosampling.....	39
Harvested Halibut Biomass	41
DISCUSSION.....	41
Creel Estimates	41
May Effort	42
Precision of the Results	46
Accuracy of the Results.....	46
RECOMMENDATIONS	47
ACKNOWLEDGEMENTS.....	48
LITERATURE CITED.....	48
APPENDICES	50
Appendix 1: Data forms	50
Appendix 2: HPUE estimates by period, trip type, subarea and species.....	53
Appendix 3: Counts of actively fishing sport vessels in each subarea, by date	63
Appendix 4: Retained and released fish by month, subarea, and species.	64
Appendix 5: Adipose fin-clip incidence.....	76

LIST OF TABLES

Table 1.	Sample sizes to estimate non-lodge angler activity patterns.....	6
Table 2.	Overflight sampling effort.....	9
Table 3.	Monthly number of shifts, and numbers of boats interviewed per landing site	14
Table 4.	Fishing locations as reported by interviewed non-lodge parties, by month and subarea	15
Table 5.	Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea	15
Table 6.	Numbers of guided and unguided boat trips, by landing site.....	15
Table 7.	Sample sizes for CPUE estimates by period, for guided and not guided non-lodge boat trips	18
Table 8.	Subarea data used to calculate CPUE in each period and subarea	18
Table 9.	Kruskal-Wallis tests for differences in HPUE among periods, by species	20
Table 10.	Paired t-tests for differences in HPUE between guided and non-guided boats, by species	21
Table 11.	Average number of harvested fish per boat trip, by month, from the census of Area 3 lodges.....	21
Table 12.	Lodge, non-lodge, and total effort estimates, by month and subarea	23
Table 13.	Total estimated harvest, by subarea, for popular sportfish taxa	29
Table 14.	Total estimated Chinook Salmon catch, by subarea	29
Table 15.	Total estimated Coho Salmon catch, by subarea	30
Table 16.	Total estimated Pink Salmon catch, by subarea	30
Table 17.	Total estimated Chum Salmon catch, by subarea	31
Table 18.	Total estimated Sockeye Salmon catch, by subarea	31
Table 19.	Total estimated Pacific Halibut catch, by subarea.....	32
Table 20.	Total estimated Lingcod catch, by subarea	32
Table 21.	Total estimated catch of other groundfish by subarea	33
Table 22.	Total estimated rockfish catch, by subarea.....	34
Table 23.	Kruskal Wallis tests for differences in harvest months, by species.....	36
Table 24.	Biosampling results.....	38
Table 25.	Average weight, and harvest of Pacific Halibut	39
Table 26.	Monthly recreational fishing effort in Area 3 and 4 from 1995 to 2024.....	44
Table 27.	Monthly Chinook Salmon, Coho Salmon, Pacific Halibut, and Lingcod harvest, and Chinook Salmon releases, in the Area 3 and 4 recreational fishery from 1995 to 2024.....	45

LIST OF FIGURES

Figure 1.	Map of the North Coast study area	2
Figure 2.	Map showing planned flight path of effort survey overflights.....	8
Figure 3.	Number of interviews conducted at each of three landing sites, by month	13
Figure 4.	Angler activity pattern by day type and month	16
Figure 5.	Average number of fish harvested per non-lodge boat trip, for six sportfish species, by period and subarea.....	19
Figure 6.	Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites.....	22
Figure 7.	Total estimated numbers of boat trips, by month and subarea	24
Figure 8.	Heat map showing the estimated total effort in each subarea	25
Figure 9.	Total estimated Chinook Salmon harvest, by month and subarea.....	35
Figure 10.	Total estimated Coho Salmon harvest, by month and subarea.....	35
Figure 11.	Total estimated Pacific Halibut harvest, by month and subarea.....	37
Figure 12.	Total estimated Lingcod harvest, by month and subarea	37
Figure 13.	Length-frequency distributions of biosampled fish.....	40
Figure 14.	Fishing effort from 1995 to 2024.....	41
Figure 15.	Harvest of Chinook and Coho salmon, Pacific Halibut, and Lingcod from 1995 to 2024	43
Figure 16.	Number of Chinook Salmon released from June to August by anglers in Areas 3 and 4 from 1995 to 2024	44

INTRODUCTION

The North Coast recreational fishery, spanning Management Areas 3 and 4 of DFO's Pacific Region, attracts resident and tourist anglers, targeting a variety of species. Predominantly focused on Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, anglers also pursue Pacific Halibut (*Hippoglossus stenolepis*), Lingcod (*Ophiodon elongates*), and various other groundfish, rockfishes (*Sebastes* spp.), and invertebrate species. Over the last ~30 years, fishing activity in this area has notably increased (Robichaud and Addison 2024), raising conservation concerns, and prompting the need to understand the harvest dynamics within the recreational fishery of Areas 3 and 4. Harvested salmon can originate from Canadian or US (enhanced and wild) stocks, hence there is a critical need for this creel program to determine BC fishery impacts, especially for stocks of concern.

This creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production. Aside from providing precise estimates of harvest and effort, the creel program facilitates direct field observations of catches, enabling verification of species identification, quantification of harvests, and collection of biological samples from Chinook and Coho salmon. Notably, crucial data such as CWT (Coded Wire Tag) information, integral to annual stock assessments and the management of Nass and Skeena river Chinook and Coho salmon stocks, are derived from the heads of randomly sampled adipose-clipped fish collected by program observers.

Since its inception in 2008, the creel survey program in Areas 3 and 4 has been instrumental in estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud and Addison 2024). In 2015, the responsibility for conducting the creel survey transitioned from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey (Robichaud et al. 2016) followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2024 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous 14 years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021, 2022, 2023, 2024), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island (Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

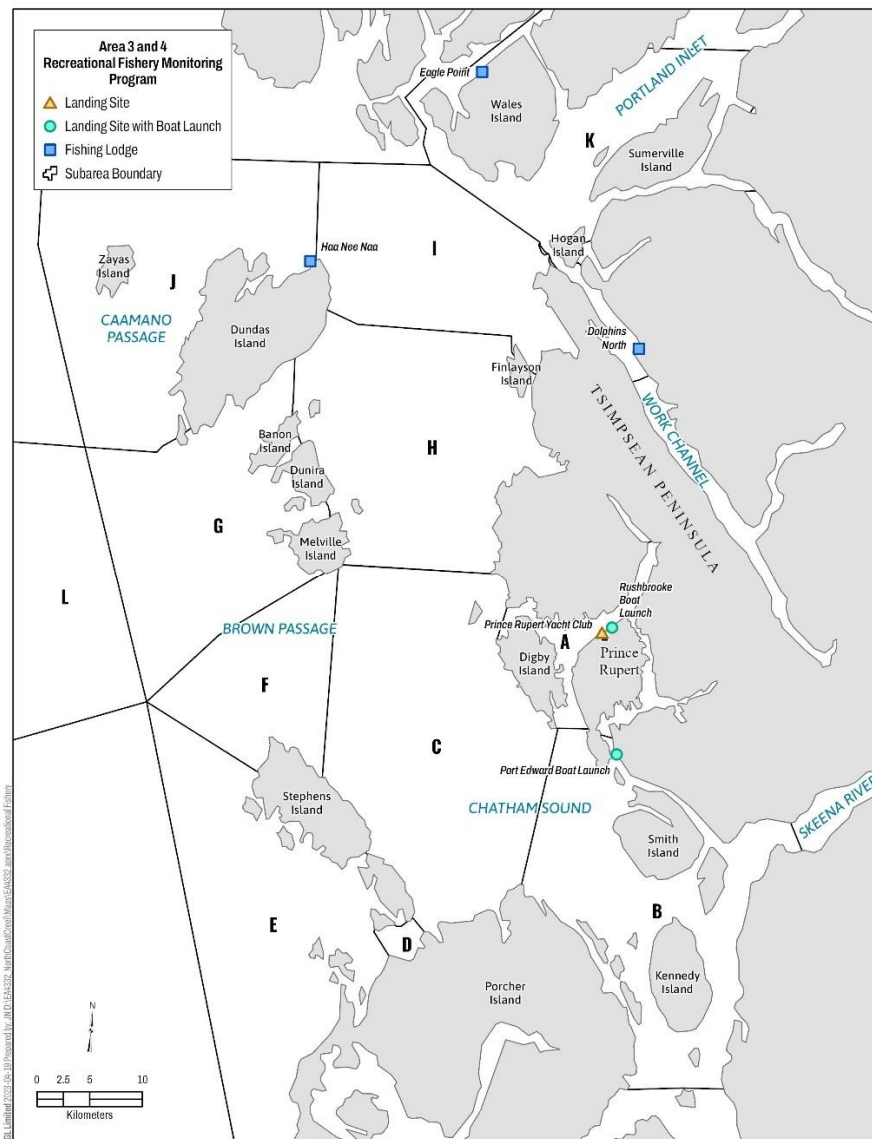


Figure 1. Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas that were used as spatial strata in the creel survey design.

The temporal scope of the full study was from 1 May through 31 August 2024. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut, and Lingcod.

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted

by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from May through August 2024. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend/holiday days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Maintain software for data management and analysis comparable to DFO's CREST program;
2. Provide monthly (May-August) estimates of the catch and effort of all salmon species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO and PSC with monthly in-season catch estimates for all salmon species harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. During angler interviews, collect biological data from a representative subsample of the landed harvest, including fin clip incidence, lengths, and scale or tissue samples for Coho and Chinook salmon;
6. Produce a final written report to be available to a technical audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests, and First Nations.

Objective 1 was addressed over the course of the study period. Objectives 2 and 6 were met by completing this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the finalization of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured for length and sexed, scale or tissue samples and adipose fin clip incidence data were collected for Coho and Chinook salmon, heads were collected from adipose-clipped fish, and all biological data were submitted to DFO at the end of study period. Objective 7 was met on 5 December 2024 at the DFO IHPC Post-Season Review meeting.

METHODS

A complete census of catch and effort was available for lodge-based anglers. The census data were collected by lodge staff and shared voluntarily with Fisheries and Oceans Canada stock assessment personnel. The remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the

number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested Chinook and Coho salmon were measured (nose to fork length), sexed, checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon, and tissue samples (fin clips) from Coho Salmon. Chinook Salmon were also examined for flesh colour. Pacific Halibut were measured (total length);
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, similar to those used in previous years (e.g., Robichaud and Addison 2024), provided a statistically unbiased estimate of catch per unit effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that is typically conducted from that site, in order to maximize the number of interviews obtained. This approach was based on two important observations: 1) the variability in CPUE (catch per unit effort) among fishing parties landing at a single access point tends to be as great as the variability in CPUE among different access points within a geographic area; and 2) CPUE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site.

During an observer shift, attempts were made to interview all anglers returning to the access location on that day, but some were missed (it is assumed that a random process dictated whether a boat was missed or interviewed), and some anglers refused to participate (it is not clear how the angling behaviour of the refusals differed from that of the participants). The geographic distribution of our samples was dictated by where the anglers fished, and it was assumed that the distribution of samples taken at the access locations was random relative to the fish harvested by the fishery.

The frequency of interview shifts was planned with the target of sampling 20% of estimated effort, while attempting to have “hands on” and direct observations of (and biological samples from) as many fish as possible meant allocating adequate time to each interview. The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Interviewing sessions were separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00, and PM shifts from 13:00 to 21:00. In total, 241 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were scheduled on 3 weekend days and 5 weekdays each month. Timing of the overflights was targeted for peak fishing periods (11 AM to 1 PM), and in 2024 most flights occurred at target times. Fog or logistical issues (base ran out of fuel) delayed the start of two flights (flight on 22 August ran from 12:14 PM to 1:45 PM; flight on 31 August ran from 12:38 PM to 2:10 PM). Analytical methods accommodate the occasional off-peak flight but can result in increased uncertainty in the effort estimate. In two months of 2025, only seven flights were flown because of weather cancellations on the last scheduled flight day (Appendix 3), coupled with the charter being unavailable for the remaining days in the month.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site a , were consistent for landing site a , even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}} \quad , \quad (\text{Equation 1})$$

where N_{md} was the total number of type d days in month m ; and K_{mda} was the number of days during which interviews occurred at landing site a , on type d days during month m .

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the recreational fishing boats that were *not* interviewed. Some recreational fishing boats were not interviewed because of refusals, and some boats returned during periods of high traffic volume when it was not possible to interview all returning vessels. Some returning vessels were not doing recreational fishing at all (e.g., pleasure cruisers, Food-Social-Ceremonial harvesters). For the recreational fishing vessels, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (u) operating on each surveying date (k) at each landing site (a):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku} \left(\frac{LF_{aku}}{LF_{aku} + LN_{aku}} \right)}{LF_{aku} - ri_{aku}} \quad . \quad (\text{Equation 2})$$

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku} , refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were

Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns, for all levels of each factor, 2024.

Month	Day Type	Subarea												TOTALS		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall
May	WD	3	6	11	104	25	39	20	2	50	6	8	10	20	264	284
	WE	1	9	5	91	43	58	26	6	53	6	9	6	15	298	313
	Total	4	15	16	195	68	97	46	8	103	12	17	16	35	562	597
June	WD	11	36	10	148	50	69	35	6	138	27	24	16	57	513	570
	WE	11	41	16	117	55	70	39	8	90	22	27	11	68	439	507
	Total	22	77	26	265	105	139	74	14	228	49	51	27	125	952	1,077
July	WD	32	48	21	173	37	103	84	6	211	31	40	9	101	694	795
	WE	20	64	10	158	55	103	56	5	198	29	52	17	94	673	767
	Total	52	112	31	331	92	206	140	11	409	60	92	26	195	1,367	1,562
August	WD	17	40	12	143	43	103	45	5	175	16	25	6	69	561	630
	WE	12	35	4	98	37	113	53	6	122	13	25	8	51	475	526
	Total	29	75	16	241	80	216	98	11	297	29	50	14	120	1,036	1,156
GRAND TOTAL		107	279	89	1032	345	658	358	44	1037	150	210	83	475	3,917	4,392

not interviewed by surveyor u , during surveying date k , at landing site a . This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term $A_{mdsakut}$ to denote the number of non-lodge boats reporting activity during time-block t , as interviewed by surveyor u on survey date k , at landing site a , in subarea s , during month m , and on day type d (n_{mdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut}) \quad . \quad (\text{Equation 3})$$

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst} \quad . \quad (\text{Equation 4})$$

The proportion of non-lodge boats (P_{mdst}) that were actively fishing during each of the hourly time-blocks (t) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad . \quad (\text{Equation 5})$$

Using this method, 96 unique angler activity patterns were to be estimated (i.e., 4 months $\times$ 2 day types $\times$ 12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 96 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subareas. The equation for non-lodge angler activity was thus:

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad , \quad (\text{Equation 6})$$

and the associated variance was:

$$S_{P_{mdt}}^2 = \frac{(P_{mdt})(1 - P_{mdt})}{\sum_s \sum_k \sum_u \sum_a (W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku})} \quad . \quad (\text{Equation 7})$$

Catch Per Unit Effort Estimation

Catch per unit effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per unit effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), month (m), and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per unit effort (CPUE), in terms of catch per boat trip, was calculated as:

$$CPUE_{msrgi} = \frac{C_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \quad (\text{Equation 8})$$

Mean $CPUE$ was calculated as the average catch, over all n_{msg} boat trips:

$$\widehat{CPUE}_{msg} = \frac{\sum_{i=1}^{n_{msg}} C_{msrgi}}{n_{msg}}. \quad (\text{Equation 9})$$

The variance for the estimate of mean catch per unit effort was calculated as:

$$S^2_{\widehat{CPUE}_{msg}} = \frac{\sum (CPUE_{msrgi} - \widehat{CPUE}_{msg})^2}{(n_{msg} - 1)}. \quad (\text{Equation 10})$$

In the event that a month/subarea-specific sample size was < 5 , adjacent subareas would be pooled together as follows:

- if n for 3I, 3J or 3K was too low, then it was combined with the other two subareas, along with data from ambiguous parts of Area 3 (“3?”) to give a 3?IJK regional average CPUE;
- if n for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPUE;
- if n for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPUE; and
- if n for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPUE.

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPUE), and the average number of fish released per boat trip (RPUE).

Angler Effort Estimation

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

Lodge Effort Census

A complete census of lodge-based angling effort was collected from lodge logbooks by DFO personnel. DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodges reported the number of active vessels for each flight date. 2) No lodge-based vessels were considered available to the

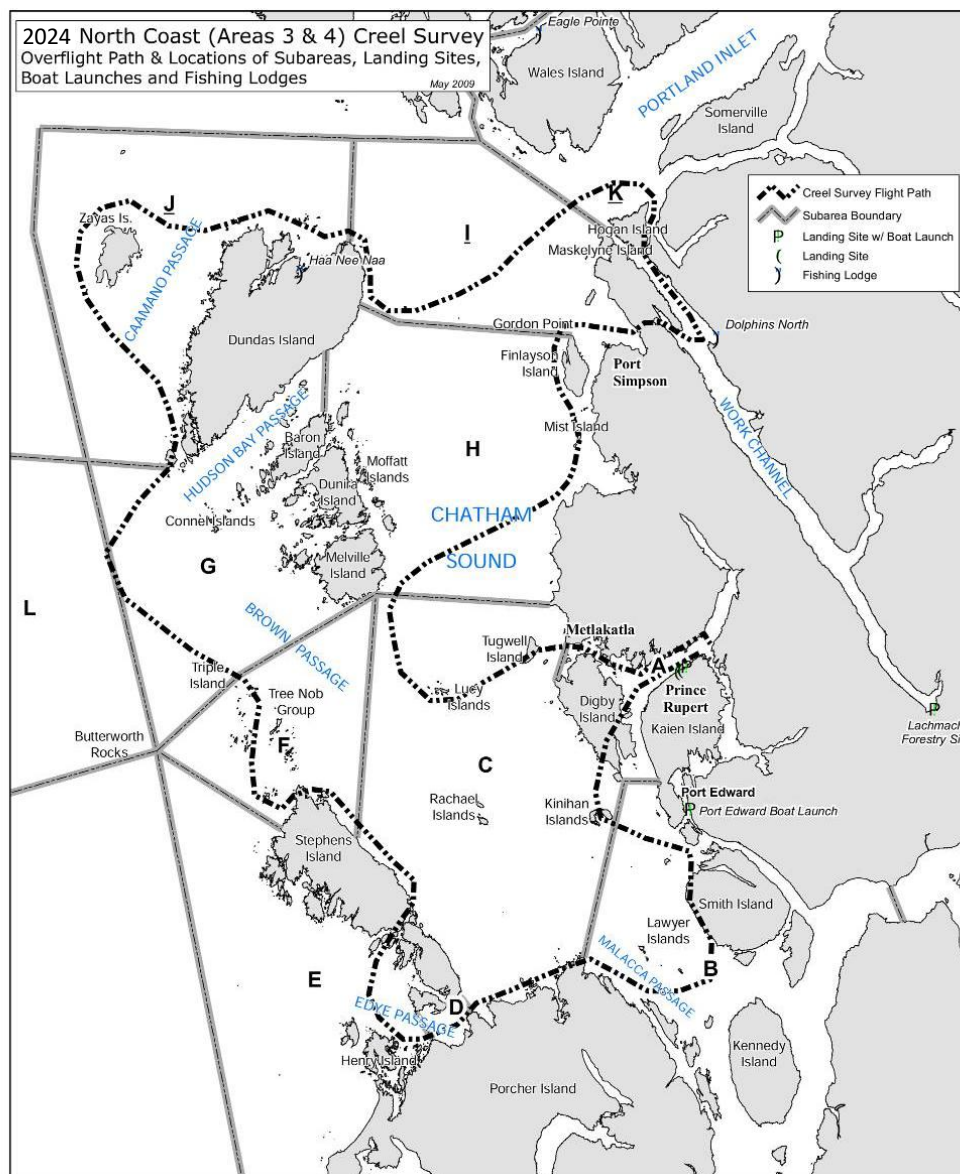


Figure 2. Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

flights (zero visible) on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight).

Change-over days were often different for each lodge. 3) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Based on this information, it was assumed that 60% of the vessel activity was available to be counted on non-changeover flight dates. 4) The number of vessels visible were summed over the active lodges.

None of the lodge effort reported from Area 3 included subarea-specific location information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

Table 2. Overflight sampling effort (number of flights shifts) by day type and month, 2024.

Month	Weekday	Weekend/	Total
		Holiday	
June	5	3	8
July	5	3	8
July	4	3	7
August	5	2	7
Total	19	11	30

Non-lodge Effort Surveys

Table 2 shows the number of overflight surveys conducted for each day type and month. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed. Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large (≥ 30 ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey o (conducted during month m and on day type d), observers tallied the total number of recreational boats that were actively fishing at time t on their i^{th} pass over subarea s , V_{mdsoit} . For each survey, the relative proportion of visible boats within Management Area 3 was calculated, ρ_{mdsoit} , and the total number of lodge boats (λ_{mdo}) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad . \quad (\text{Equation 11})$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = \max \{0, V_{mdsoit} - \lambda_{mdsoit}\} \quad . \quad (\text{Equation 12})$$

Since angling occurs over the course of the entire day, the numbers of non-lodge boats were divided by the proportion of average daily number of boats active (P_{mdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{mdso} :

$$B_{mdso} = \sum_i \frac{v_{mdsoit}}{P_{mdst}} \quad . \quad (\text{Equation 13})$$

The variance of B_{mdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} \text{if } z = x/y, \text{Var}(z) &= (y^{-2})\text{Var}(x) + (x^2 y^{-4})\text{Var}(y) ; \\ \text{if } z = xy, \text{Var}(z) &= (y^2)\text{Var}(x) + (x^2)\text{Var}(y) ; \text{ and} \\ \text{if } z = qx, \text{Var}(z) &= (q^2)\text{Var}(x) \quad \text{where } q \text{ is a constant.} \end{aligned} \quad (\text{Equation 14})$$

Since there was no variance associated with the boat counts, it followed that

$$S_{B_{mdso}}^2 = v_{mdsoit}^2 \cdot (P_{mdst})^{-4} \cdot S_{P_{mdst}}^2 \quad . \quad (\text{Equation 15})$$

Next, the survey-specific ‘daily effort’ estimates were averaged over the number of surveys conducted, n_{mds} , as:

$$\hat{B}_{mds} = \frac{\sum_{o=1}^{n_{mds}} B_{mdso}}{n_{mds}} \quad (\text{Equation 16})$$

with a variance of:

$$S_{\hat{B}_{mds}}^2 = (n_{mds})^{-2} \cdot S_{B_{mdso}}^2 \quad . \quad (\text{Equation 17})$$

Total fishing effort for each month was calculated for each subarea by multiplying the month-specific average daily effort by the number days of day type d that occurred in month m :

$$E_{mds} = \hat{B}_{mds} \cdot N_{md} \quad , \quad (\text{Equation 18})$$

which was then summed over day type to give:

$$E_{ms} = \sum_d \hat{B}_{mds} \cdot N_{md} \quad , \quad (\text{Equation 19})$$

with a variance of:

$$S_{E_{ms}}^2 = \sum_d S_{\hat{B}_{mds}}^2 \cdot N_{md}^2 \quad . \quad (\text{Equation 20})$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPUE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

Rarely, this method produced an effort estimate of zero in a stratum for which interview data suggested otherwise. In 2024, no boats were counted in Subarea 4L in May and June weekends, or in Subarea 4E in May weekends, but interviews showed that effort had occurred. In these cases, the interview data obtained were taken as representative of total effort.

Catch and Harvest Estimation

Total catch was calculated for each month, subarea, and species by multiplying month-specific angling effort by month-specific catch per unit effort, and then summing over day type:

$$C_{msr} = \sum_d (E_{mds} \cdot \widehat{CPUE}_{msg}) \quad , \quad (\text{Equation 21})$$

where $g = \text{'all'}$ (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left(E_{mds}^2 \frac{S_{\widehat{CPUE}_{msg}}^2}{n_{msg}} + \widehat{CPUE}_{msg}^2 \frac{S_{E_{mds}}^2}{n_{mds}} + \frac{S_{\widehat{CPUE}_{msg}}^2 S_{E_{mds}}^2}{n_{msg} n_{mds}} \right)} \quad , \quad (\text{Equation 22})$$

where n_{mds} and n_{msg} are the sample sizes for E and CPUE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released by replacing CPUE with HPUE or with RPUE in Equations 21 and 22.

Biosampling

When given permission by the anglers, surveyors collected biological data from Chinook and Coho salmon and Pacific Halibut. Biosampled species were sexed when they were intact. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and when possible, heads from clipped fish were collected. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon, and tissue samples (fin clips) were collected from Coho Salmon (used by DFO to determine age, and to carry out genetic stock identification).

Ideally, all anglers would allow their catch to be biosampled, and the examined fish would represent an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had surveyors ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921 \times 10^{-6} \times \text{FL(cm)}^{3.24} \\ \text{Weight(lb)}_{\text{round}} &= 9.166 \times 10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned}\quad (\text{Equation 23})$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

For each fish taxa, differences in CPUE among months were examined using Kruskal Wallis tests (Zar 1984), whereas differences between trip-types (guided vs. non-guided) were examined using paired t-tests (i.e., the difference between guided vs unguided catch for a given subarea and month was tested against a null of zero; Sokal and Rohlf 1995). Kruskal Wallis tests were used to examine the effects of month and day type on effort, and the effects of month on catch and harvest. When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Trailer Census

The total number of parked trailers was recorded daily at the landing locations during the peak of the activity timing window (i.e., between 11:00 AM and 1:30 PM; similar to the targeted timing of aerial overflights). Theoretically, trailer counts could be used as an index of fishing effort, are more economical than overflights, and are not weather dependant, though no sub-area-specific effort data are collected. DFO has been collecting trailer count data for years and we are maintaining the time series. Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. For the regression analysis, the location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

In addition to every day during the May through August study period, trailer count data were also collected throughout April. The above linear regression was used to calculate from the daily trailer counts what the daily boat counts *would have been* had there been overflights on each day. Day type-specific activity pattern data were used to expand the daily estimated ‘boat counts’ to

calculate total daily fishery effort (May activity data were used to expand both the April and May trailer count numbers, otherwise month-specific activity pattern data were used). Summing over all days in each month produced an *index* of fishing effort for each month. A scaler was calculated as the ratio of the April index value to the sum of the May to August indices. Next, that May to August creel-based non-lodge effort values were multiplied by the scaler to estimate April non-lodge fishing effort. The same analysis was repeated using trailer count data from three other years (i.e., 2019, 2020, and 2023) in which April trailer count data were collected (using the regression line and the activity pattern data from those years, see Robichaud and Addison 2020, 2021, 2024).

RESULTS

Angler Interviews

Over the four month study period, there were 247 observer shifts, during which 2,808 interviews were conducted, involving 9,041 non-lodge anglers (Figure 3; Table 3). July was the busiest month, with 996 interviews conducted (average 15.6 interviews per shift), which represented 35% of the total number of study period interviews (Table 3).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4F being the favourite (Table 4). Subareas 4A and 4C were the next most likely area for anglers to have reported activity, followed by 4B and 4D. The subareas with the lowest number of trips reporting activity were 4E and 4L. Only 108 interviews were of parties that only targeted shellfish, and those were overwhelmingly in subarea 4A (Table 5).

On average, 33% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 53 of 751 trips (7.1%) interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 34% and 52% of the time, respectively (Table 6).

In all there were 166 refusals (Table 4), or 5.9% refusal rate. The highest refusal rate (7.1%) was at the Rushbrooke boat launch, where 50% of the refusals occurred. The refusal rates for the Port Edward boat launch and the Prince Rupert Yacht Club were 3.3% and 6.5% (15% and 35% of the refusals), respectively.

Angler Activity Patterns

Non-lodge angler activity pattern estimates varied little among periods (Figure 4). There was virtually no difference between the weekday and weekend/holiday activity patterns, except for a slightly greater tendency for weekend anglers to be active during the peak times in May, as compared to weekday anglers. Timing of our overflights lined up with peak activity levels for all flights.

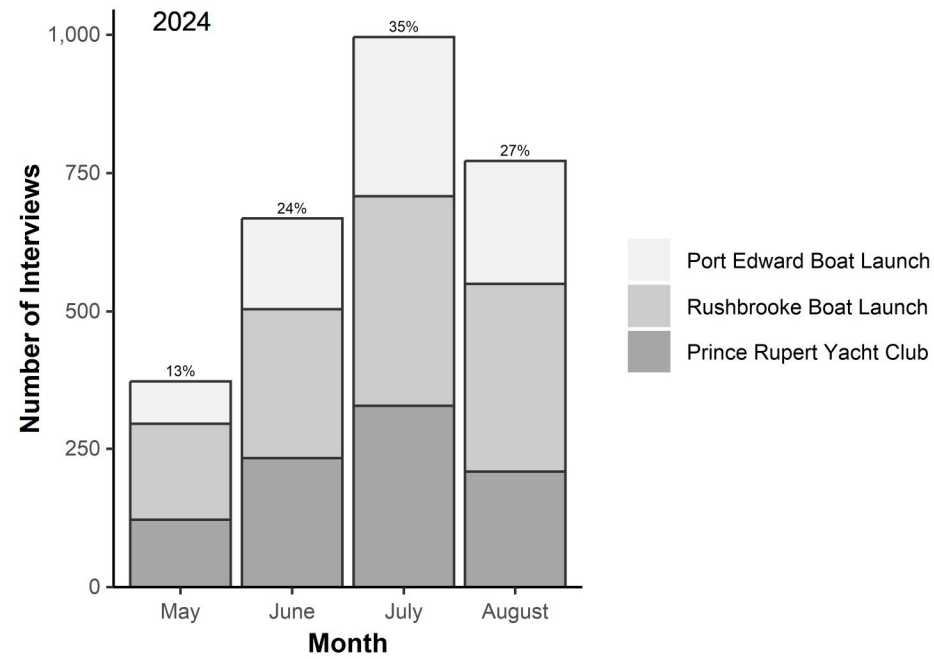


Figure 3. Number of interviews conducted at each of three landing sites, by month, 2024.

Table 3. Monthly number of shifts, and numbers of boats interviewed per landing site, 2024. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.

Landing Site	May				June				Jul			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	77	184	20	3.9	164	432	22	7.5	288	930	23	12.5
Prince Rupert Yacht Club	122	399	22	5.5	233	817	22	10.6	328	1140	22	14.9
Rushbrooke Boat Launch	173	486	21	8.2	271	796	20	13.6	380	1264	19	20.0
TOTAL	372	1,069	63	5.9	668	2,045	64	10.4	996	3,334	64	15.6
% of Total Interviews			13%				24%				35%	
Estimated Non-lodge Effort			1,685				2,779				3,614	
% of Cumulative Non-lodge Effort			14%				23%				30%	
% of Effort Sampled			22%				24%				28%	

Landing Site	August				TOTAL			
	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	222	674	20	11.1	751	2220	85	8.8
Prince Rupert Yacht Club	209	786	18	11.6	892	3142	84	10.6
Rushbrooke Boat Launch	341	1133	18	18.9	1165	3679	78	14.9
TOTAL	772	2,593	56	13.8	2,808	9,041	247	11.4
% of Total Interviews			27%					
Estimated Non-lodge Effort			3,850				11,928	
% of Cumulative Non-lodge Effort			32%					
% of Effort Sampled			20%				24%	

Table 4. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2024. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?	Refusal	
May	2	9	10	104	45	65	30	4	62	6	9	7		1	21	372
June	11	47	17	133	72	93	49	8	139	28	29	12	1	3	29	668
July	27	69	20	166	67	127	106	7	266	28	46	13	1	3	51	996
August	17	48	11	121	56	135	69	7	187	14	30	7	1	3	65	772
TOTAL	58	173	57	524	240	420	253	25	654	75	113	39	2	10	166	2,808

Table 5. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2024.

Month	Subarea															TOTAL
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	3?	4?		
May	0	0	1	19	0	0	0	0	0	0	0	0	0	0	0	20
June	0	0	0	23	1	0	0	0	0	0	0	0	0	0	0	24
July	0	0	0	34	1	2	0	0	0	1	1	0	0	0	0	38
August	0	0	0	21	2	2	0	0	1	0	0	0	0	0	0	26
TOTAL	0	0	1	97	4	4	0	0	1	1	1	0	0	0	0	108

Table 6. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2024.

Landing Site	Trip Type		% guided
	Guided	Unguided	
Port Edward Boat Launch	53	698	7.1%
Prince Rupert Yacht Club	468	424	52%
Rushbrooke Boat Launch	394	771	34%
TOTAL	915	1,893	33%

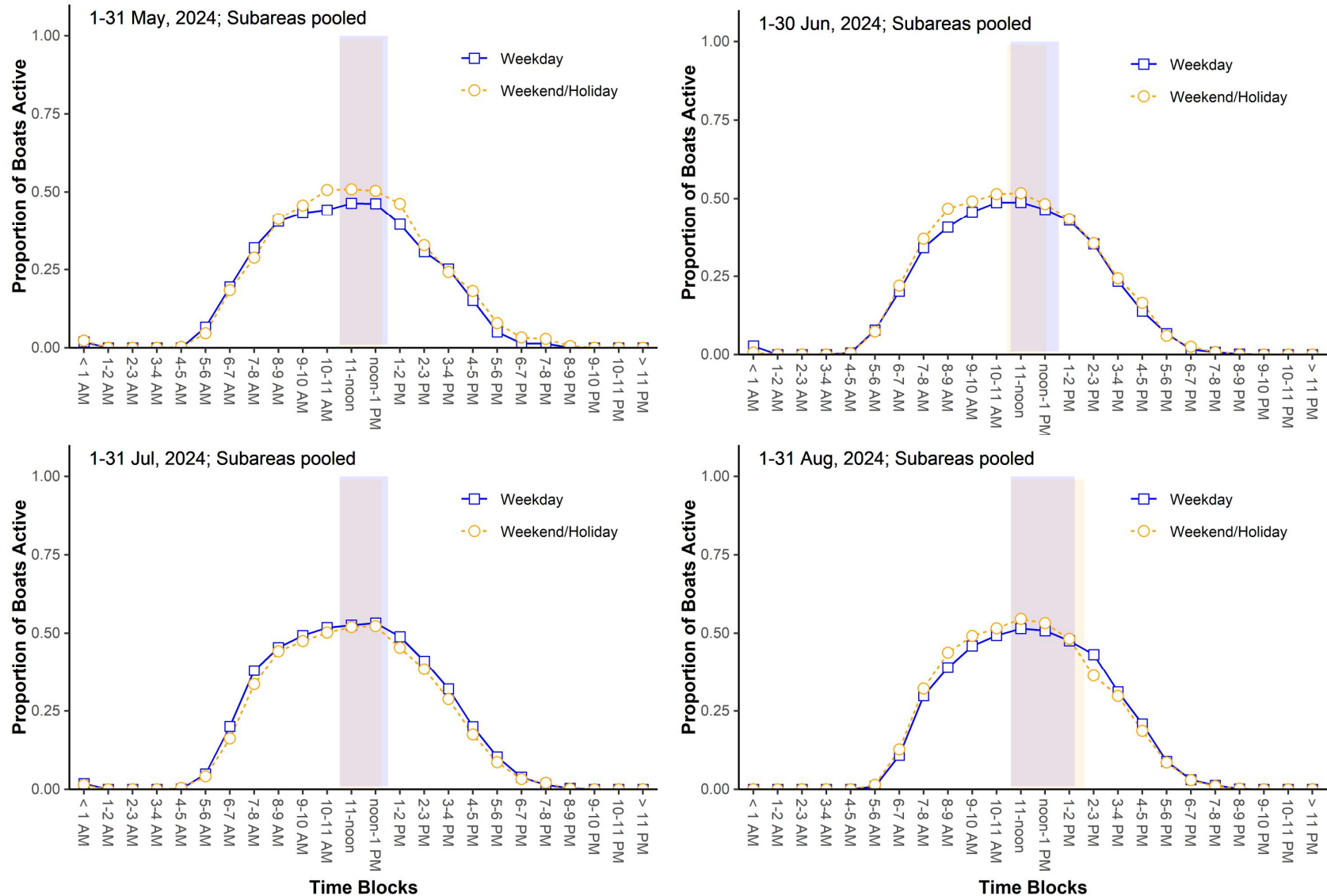


Figure 4. Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas, 2024. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and period.

Catch Per Unit Effort Estimates

Sample sizes were adequate ($n \geq 5$) to produce reliable estimates of non-lodge CPUE in most combinations of periods and creel study sub-areas (Table 7), except in subarea 3I in May. However, when the interview data were split between guided and non-guided groups, period-specific sample sizes did not suffice for several more combinations of period and management area (Table 7). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPUE calculations (Table 8).

Average number of fish harvested per non-lodge boat trip (HPUE) was calculated for each species by period, subarea, and by trip type (Appendix 2).

HPUE varied significantly among periods (Figure 5, Table 9) for Chinook ($\chi^2 = 9.4$, $df = 3$, $P = 0.025$), Coho ($\chi^2 = 21.8$, $df = 3$, $P = 0.0001$) and Pink salmon (*Oncorhynchus gorbuscha*, $\chi^2 = 19.0$, $df = 3$, $P = 0.0003$). Chinook Salmon HPUE was highest in May, declined into June, and was lower still in July and August, but these among-month differences were not statistically significant after controlling for the rate of false positives (Table 9). For Coho Salmon, median HPUE increased from May to July, and declined thereafter (Table 9; Figure 5), with May being significantly lower than all other months, and with June being significantly lower than July. HPUE for Pink Salmon was significantly lower in May and June as compared to July and August. HPUE for Tiger Rockfish (*Sebastes nigrocinctus*) also varied significantly among periods (Table 9), but the among period differences were not statistically significant after controlling for the rate of false positives. There were no significant differences for any other species examined (including Pacific Halibut and Lingcod; Figure 5).

Guided trips typically had higher HPUE than non-guided trips, and the differences were statistically significant for a variety of species (Table 10). In particular, guided trips harvested 0.5 more Chinook Salmon ($t_{47} = 4.9$, $P < 0.0001$), and 0.7 more Coho Salmon ($t_{47} = 2.9$, $P = 0.005$) than unguided trips. In addition, guided trips harvested 0.4 more Lingcod ($t_{47} = 3.4$, $P = 0.001$), 0.2 more Pacific Cod (*Gadus macrocephalus*, $t_{47} = 2.7$, $P = 0.009$), and 0.7 more Pacific Halibut per trip ($t_{47} = 5.0$, $P < 0.0001$) than unguided trips. Guided trips also harvested significantly more Quillback Rockfish (*Sebastes maliger*) than unguided trips.

For lodge boat trips, HPUE (and RPUE for Chinook Salmon) values are shown in Table 11.

Table 7. Sample sizes for CPUE (catch per unit effort) estimates by period, for guided and not guided non-lodge boat trips, 2024.

		Subarea											
Period	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	1	7	6	99	3	27	14	4	52	5	6	12
	Unguided	3	8	10	96	65	70	32	4	51	7	11	4
	Combined	4	15	16	195	68	97	46	8	103	12	17	16
June	Guided	11	42	9	169	6	33	15	3	113	24	25	23
	Unguided	11	35	17	96	99	106	59	11	115	25	26	4
	Combined	22	77	26	265	105	139	74	14	228	49	51	27
July	Guided	27	60	8	189	6	58	25	8	167	25	32	18
	Unguided	25	52	23	142	86	148	115	3	242	35	60	8
	Combined	52	112	31	331	92	206	140	11	409	60	92	26
August	Guided	12	41	2	142	5	58	13	2	140	11	19	9
	Unguided	17	34	14	99	75	158	85	9	157	18	31	5
	Combined	29	75	16	241	80	216	98	11	297	29	50	14

Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPUE in each period and subarea, for guided and not guided non-lodge boat trips, 2024.

		Subarea											
Period	Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
May	Guided	3?,I,J,K	3J	3K	4A	4A,B,C	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3?,I,J,K	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3?,I,J,K	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June	Guided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
July	Guided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August	Guided	3I	3J	3?,I,J,K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L

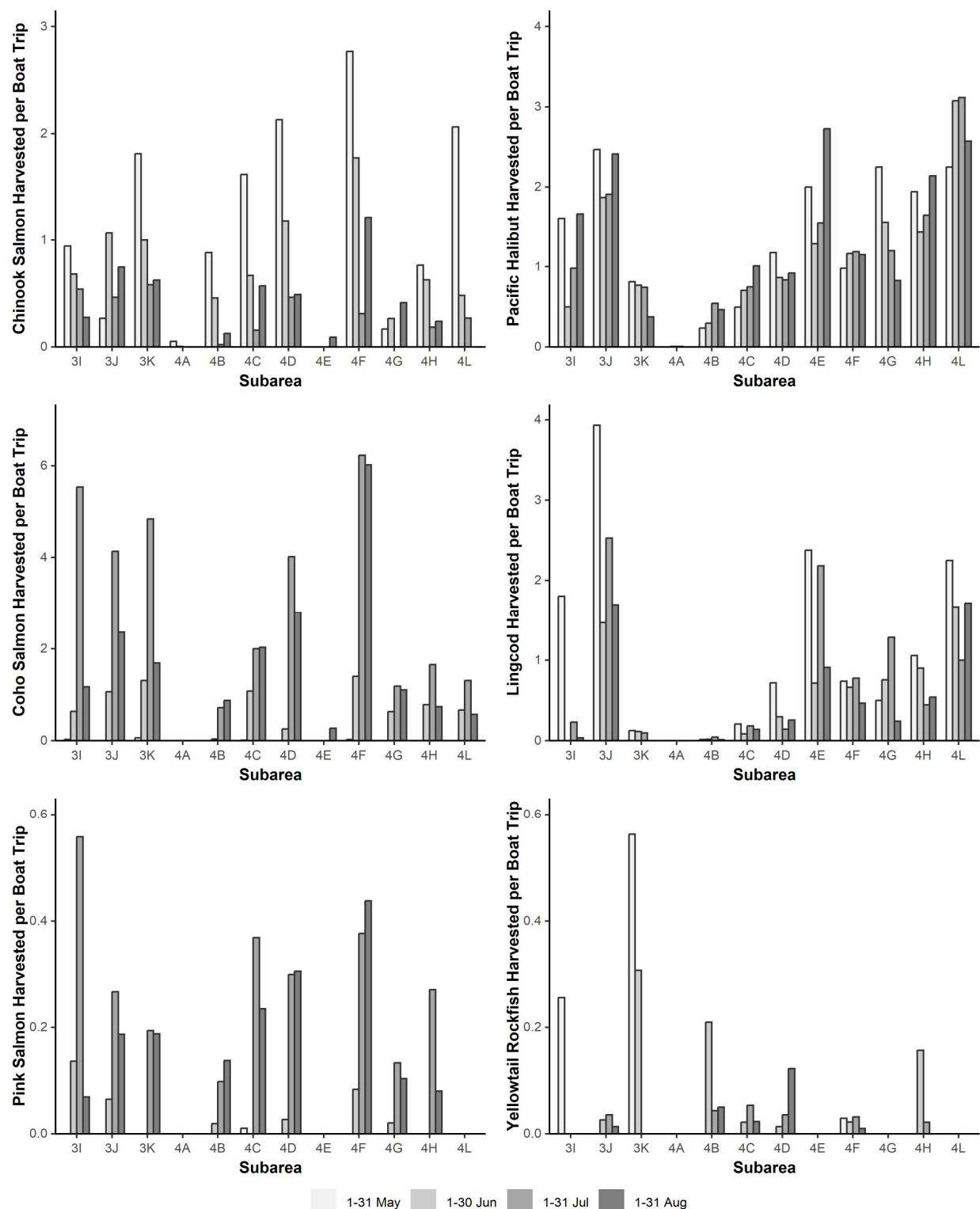


Figure 5. Average number of fish harvested per non-lodge boat trip, for six sportfish species, by period and subarea, 2024. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

Table 9. Kruskal-Wallis tests for differences in HPUE (fish harvested per non-lodge boat trip) among periods, by species, 2024. Subareas treated as replicates. Red letters indicate statistical differences within species (periods not connected by a common letter were statistically different).

Species		Kruskal-Wallis Statistics			Median HPE			
		χ^2	df	P	May	June	July	August
Chinook Salmon	<i>Oncorhynchus tshawytscha</i>	9.4	3	0.025	A 0.9	A 0.6	A 0.2	A 0.3
Chum Salmon	<i>Oncorhynchus keta</i>	2.2	3	0.52	0	0	0	0
Coho Salmon	<i>Oncorhynchus kisutch</i>	21.8	3	0.0001	C 0	B 0.7	A 1.8	AB 1.1
Pink Salmon	<i>Oncorhynchus gorbuscha</i>	19.0	3	0.0003	C 0	B 0.010	A 0.2	A 0.1
Sockeye Salmon	<i>Oncorhynchus nerka</i>	3.4	3	0.33	0	0	0	0
Arrow-tooth Flounder	<i>Atheresthes stomias</i>	6.7	3	0.08	0	0	0	0
Big Skate	<i>Beringraja binoculata</i>	2.0	3	0.56	0	0	0	0
Cabezon	<i>Scorpaenichthys marmoratus</i>	4.4	3	0.22	0	0	0	0
English Sole	<i>Parophrys vetulus</i>	2.0	3	0.58	0	0	0	0
Greenlings	<i>Hexagrammos</i> spp.	3.4	3	0.33	0.01	0	0	0
Lingcod	<i>Ophiodon elongatus</i>	2.5	3	0.48	0.7	0.5	0.3	0.2
Longnose Skate	<i>Caliraja rhina</i>	3.0	3	0.39	0	0	0	0
Pacific Cod	<i>Gadus macrocephalus</i>	5.6	3	0.13	0.1	0.1	0.04	0.3
Pacific Halibut	<i>Hippoglossus stenolepis</i>	0.7	3	0.87	1.4	1.0	1.1	1.1
Pacific Rock Sole	<i>Lepidopsetta bilineata</i>	2.6	3	0.47	0	0	0	0
Pacific Spiny Dogfish	<i>Squalus suckleyi</i>	4.3	3	0.23	0	0	0	0
Sablefish	<i>Anoplopoma fimbria</i>	3.0	3	0.39	0	0	0	0
Spotted Ratfish	<i>Hydrolagus colliei</i>	0.0	3	1	0	0	0	0
Starry Flounder	<i>Platichthys stellatus</i>	2.0	3	0.58	0	0	0	0
Black Rockfish	<i>Sebastes melanops</i>	2.0	3	0.56	0.2	0.1	0.2	0.1
Boccacio Rockfish	<i>Sebastes paucispinis</i>	0.9	3	0.83	0	0	0	0
Canary Rockfish	<i>Sebastes pinniger</i>	2.7	3	0.44	0	0	0	0
China Rockfish	<i>Sebastes nebulosus</i>	6.9	3	0.07	0.06	0	0.004	0
Copper Rockfish	<i>Sebastes caurinus</i>	4.0	3	0.26	0.03	0	0.004	0
Dusky Rockfish	<i>Sebastes ciliatus</i>	0.7	3	0.88	0	0	0	0
Quillback Rockfish	<i>Sebastes maliger</i>	0.8	3	0.86	0.10	0.09	0.1	0.09
Redbanded Rockfish	<i>Sebastes babcocki</i>	0.0	3	1	0	0	0	0
Redstripe Rockfish	<i>Sebastes proriger</i>	6.1	3	0.11	0	0	0	0
Tiger Rockfish	<i>Sebastes nigrocinctus</i>	9.5	3	0.023	A 0	A 0	A 0.02	A 0.005
Widow Rockfish	<i>Sebastes entomelas</i>	2.0	3	0.58	0	0	0	0
Yelloweye Rockfish	<i>Sebastes ruberrimus</i>	2.3	3	0.51	0	0	0	0
Yellowtail Rockfish	<i>Sebastes flavidus</i>	2.0	3	0.57	0	0.02	0.01	0

Table 10. Paired t-tests for differences in HPUE (fish harvested per non-lodge boat trip) between guided and non-guided boats, by species, 2024. Periods and subareas treated as replicates.

<i>Species</i>	<i>Paired t-test Statistics</i>			<i>Median HPE</i>		<i>Paired Mean Difference</i>
	<i>t</i>	<i>df</i>	<i>P</i>	<i>Guided</i>	<i>Not</i>	
Chinook Salmon	4.9	47	< 0.0001	0.6	0.3	0.5
Chum Salmon	1.2	47	0.25	0	0	0.005
Coho Salmon	2.9	47	0.005	1.0	0.6	0.7
Pink Salmon	-0.6	47	0.55	0	0.02	-0.0118
Sockeye Salmon	-0.6	47	0.54	0	0	-0.0009
Arrow-tooth Flounder	-0.7	47	0.50	0	0	-0.0004
Big Skate	0.4	47	0.70	0	0	0.0002
Cabezon	-1.3	47	0.19	0	0	-0.0048
English Sole	-1.2	47	0.24	0	0	-0.0014
Greenlings	0.4	47	0.69	0	0	0.005
Lingcod	3.4	47	0.001	0.6	0.3	0.4
Longnose Skate	1.4	47	0.16	0	0	0.0007
Pacific Cod	2.7	47	0.009	0.06	0.06	0.2
Pacific Halibut	5.0	47	< 0.0001	1.3	0.9	0.7
Pacific Rock Sole	1.0	47	0.33	0	0	0.01
Pacific Spiny Dogfish	-1.6	47	0.13	0	0	-0.0014
Sablefish	-1.0	47	0.32	0	0	-0.0001
Spotted Ratfish	0.0	47	1	0	0	0
Starry Flounder	-1.8	47	0.07	0	0	-0.0016
Black Rockfish	0.7	47	0.52	0.08	0.1	0.08
Boccacio Rockfish	1.4	47	0.16	0	0	0.01
Canary Rockfish	0.2	47	0.86	0	0	0.002
China Rockfish	-0.8	47	0.45	0	0	-0.0072
Copper Rockfish	1.8	47	0.07	0	0	0.03
Dusky Rockfish	-1.1	47	0.27	0	0	-0.0039
Quillback Rockfish	2.1	47	0.037	0.1	0.09	0.06
Redbanded Rockfish	0.0	47	1	0	0	0
Redstripe Rockfish	-1.4	47	0.16	0	0	-0.0026
Tiger Rockfish	1.8	47	0.07	0	0	0.02
Widow Rockfish	-2.0	47	0.06	0	0	-0.0018
Yelloweye Rockfish	-1.2	47	0.26	0	0	-0.0129
Yellowtail Rockfish	1.8	47	0.08	0	0	0.08

Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, by month, from the census of Area 3 lodges, 2024.

<i>Species</i>	<i>May</i>	<i>June</i>	<i>July</i>	<i>August</i>	<i>Total</i>
<i>Chinook Salmon:</i>					
<i>retained</i>	1.0	0.7	0.3	0.4	0.5
<i>released</i>	0.07	1.0	1.3	0.5	0.9
<i>Coho Salmon</i>	0.04	0.8	3.6	3.1	2.3
<i>Pink Salmon</i>	0	0.003	0.03	0.02	0.02
<i>Chum Salmon</i>	0	0	0	0	0
<i>Sockeye Salmon</i>	0	0	0	0	0
<i>Pacific Halibut</i>	0.5	0.6	0.7	0.6	0.6
<i>Lingcod</i>	1.0	0.8	1.0	0.6	0.8
<i>Other groundfish</i>	0	0	0	0	0
<i>Yelloweye Rockfish</i>	0	0	0	0	0
<i>Other rockfish</i>	0.6	0.2	0.4	0.2	0.3

Angler Effort Estimates

Over the four month study period, a total of 30 effort surveys were conducted, including 19 weekday surveys and 11 conducted on weekends/holidays (Table 2, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts (“the non-lodge small boat counts”) were significantly correlated to trailer counts at both survey launch sites (Rushbrooke: $F_{1,28} = 124.3$, $P < 0.0001$; Port Edwards: $F_{1,28} = 53.4$, $P < 0.0001$), and when the two launch sites were combined ($F_{1,28} = 128.6$, $P < 0.0001$). The fit of the Port Edward regression (Port Edward $R^2 = 0.64$) was lower than that of the Rushbrooke regression ($R^2 = 0.81$) or that of combined-site regression ($R^2 = 0.81$; Figure 6).

Monthly angling effort for each subarea in 2024 is shown in Table 12 and in Figure 7, and for all months combined in Figure 8. Total lodge effort in 2024 was 1,801 boat trips (166 in May, 556 in June, 633 in July, and 446 in August; Table 12). To calculate total (lodge + non-lodge) fishing effort by subarea, lodge boat trips were distributed among the three subareas in Management Area 3 according to the relative non-lodge effort in each (Table 12; Figure 7). In all, there

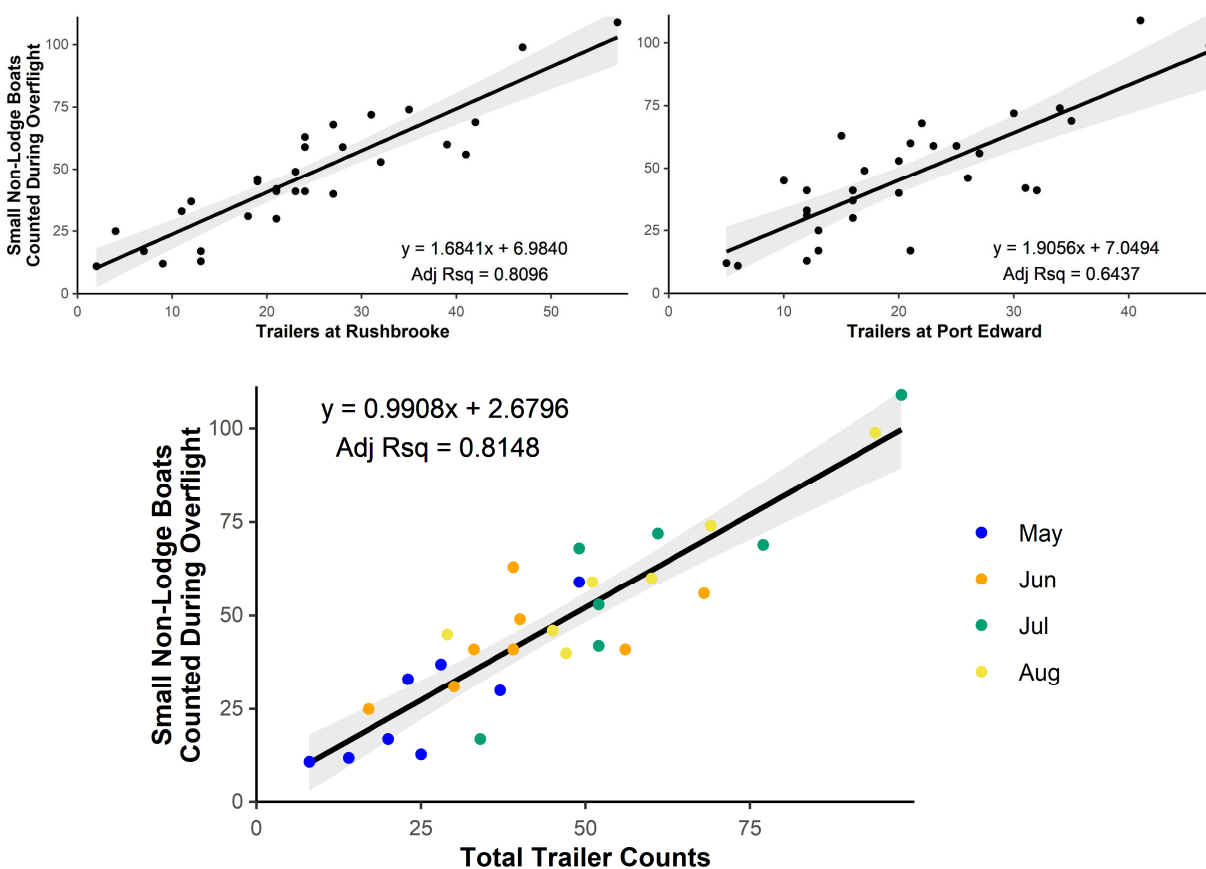


Figure 6. Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded areas show the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.

Table 12. Lodge, non-lodge, and total effort estimates (boat trips per month), by month and subarea, 2024.

Month		Subarea												TOTALS			SE of the Totals		
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May	Creel Estimate	71.4	95.7	80.1	86.8	261.5	355.3	95.9	62.9	360.9	87.7	73.8	53.4	247	1,438	1,685	4	12	12
	Lodge Report	48.0	64.3	53.8										166	0	166			
	TOTAL MAY	119.4	160.0	133.8	86.8	261.5	355.3	95.9	62.9	360.9	87.7	73.8	53.4	413	1,438	1,851	4	12	12
	SE	1.6	2.6	1.7	1.4	5.5	5.1	2.2	1.6	7.6	2.1	1.8	2.2						
June	Creel Estimate	141.9	228.1	201.6	98.1	201.4	370.5	224.3	78.4	629.1	446.4	82.7	76.6	572	2,207	2,779	5	12	13
	Lodge Report	138.1	221.9	196.1										556	0	556			
	TOTAL JUNE	280.0	450.0	397.7	98.1	201.4	370.5	224.3	78.4	629.1	446.4	82.7	76.6	1,128	2,207	3,335	5	12	13
	SE	2.1	3.4	3.2	1.5	3.3	3.7	3.2	1.2	8.4	6.3	1.4	1.1						
July	Creel Estimate	229.8	413.7	277.5	89.8	216.7	630.6	304.0	39.3	918.7	238.3	229.5	26.7	921	2,693	3,614	7	14	16
	Lodge Report	157.9	284.3	190.8										633	0	633			
	TOTAL JULY	387.7	698.0	468.3	89.8	216.7	630.6	304.0	39.3	918.7	238.3	229.5	26.7	1,554	2,693	4,247	7	14	16
	SE	2.5	5.9	3.4	1.2	2.3	5.7	4.2	0.7	10.9	4.0	3.0	0.5						
August	Creel Estimate	277.8	622.0	296.9	75.8	187.5	736.0	204.4	63.7	703.1	322.0	336.3	24.0	1,197	2,653	3,850	10	13	16
	Lodge Report	103.5	231.8	110.6										446	0	446			
	TOTAL AUGUST	381.4	853.8	407.6	75.8	187.5	736.0	204.4	63.7	703.1	322.0	336.3	24.0	1,643	2,653	4,296	10	13	16
	SE	3.3	8.3	4.2	1.0	2.3	6.5	3.1	1.3	8.5	3.5	4.0	0.5						
FULL SEASON TOTAL		1,168	2,162	1,407	350	867	2,092	829	244	2,612	1,094	722	181	4,738	8,992	13,729	14	25	29
FULL SEASON SE		5	11	6	3	7	11	6	3	18	9	6	3						

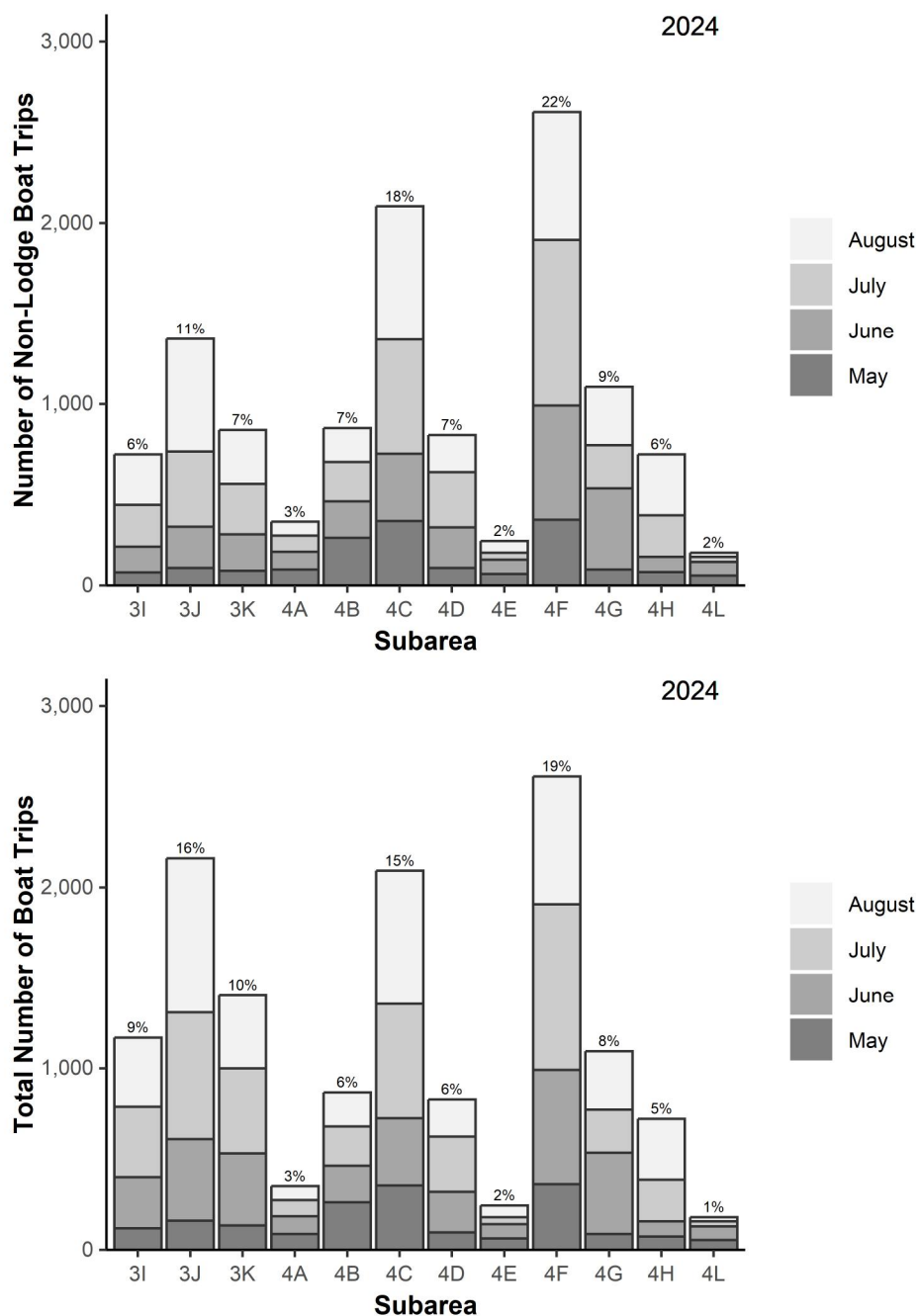


Figure 7. Total estimated numbers of boat trips, by month and subarea, 2024. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

were 1,851 (SE = 12) boat trips in May, 3,335 (SE = 13) boat trips in June, 4,247 (SE = 16) boat trips in July, and 4,296 (SE = 16) boat trips in August. The full-season (May-August) Area 3-4 effort estimate for 2024 was 13,729 boat trips (SE = 29). For comparison with our previous study years, which did not include surveys in May, the June-August effort total was 11,878 boat trips (SE = 26).

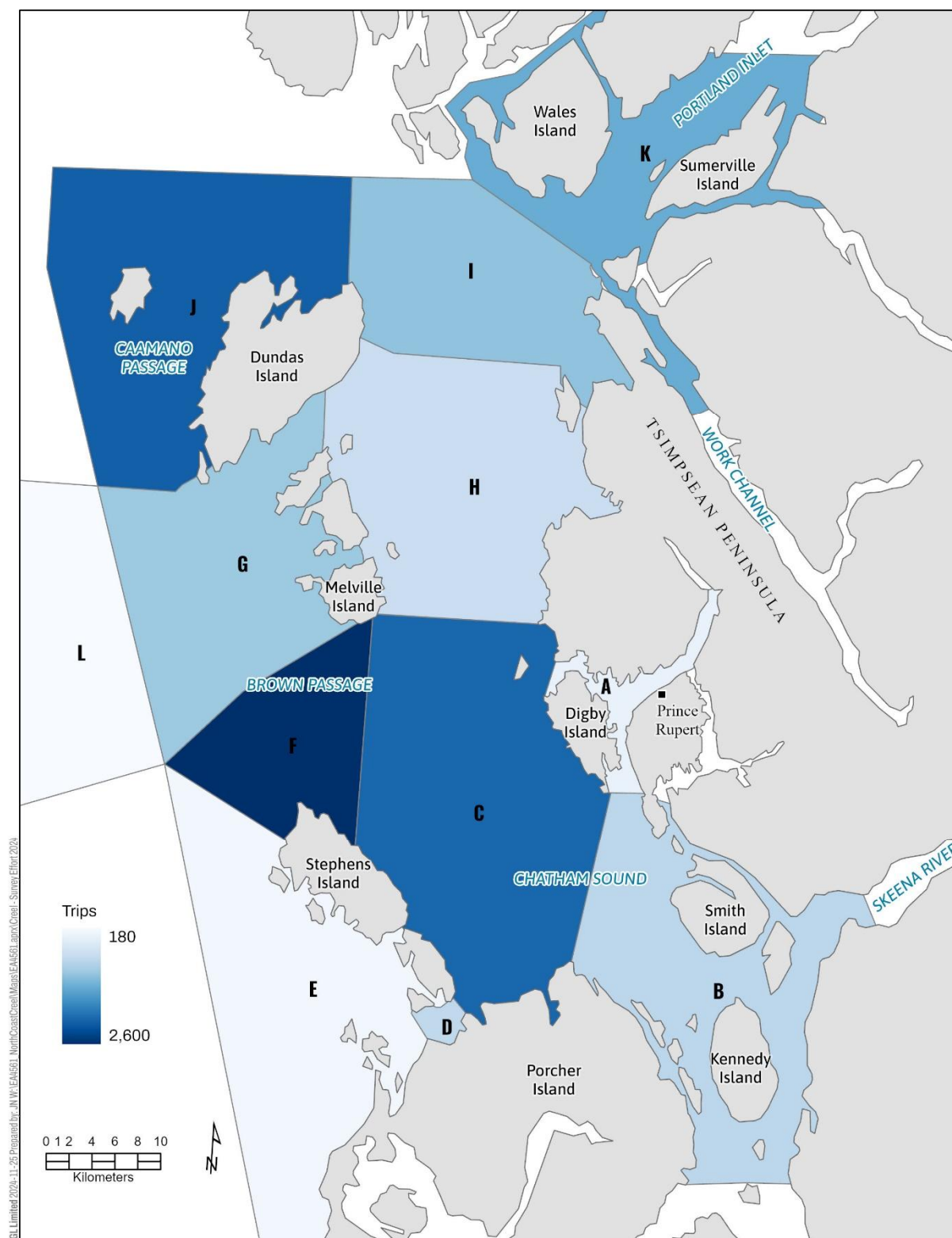


Figure 8. Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2024. See Table 12 for details.

Median daily non-lodge effort on weekdays (5.7 trips per day per subarea) was lower than on weekends/holidays (7.4 trips per day per subarea) but the effect was not statistically significant ($\chi^2 = 1.0$, $df = 1$, $P = 0.31$). Effort varied significantly among months ($\chi^2 = 10.6$, $df = 3$, $P = 0.014$). Effort in May (3.1 trips per day per subarea) was not significantly different from that in June (6.2) but was significantly lower than during July and August (8.5 and 9.1 trips per day).

April Effort

In 2019, 2020, 2023, and 2024, trailer counts began in April, and ran until the end of the creel monitoring period. The annual trailer-to-boat-count regression line (see Figure 6 for 2024) was used to calculate what the daily boat counts would have been had there been overflights each day, and the proportion of active anglers (specific to year, month, day type, and time of trailer count, see Figure 4 for 2024) was used to expand the boat counts to calculate total fishery effort by day. Summing over all the days of each month produced trailer count derived monthly fishing effort indices (Figure 9), which compared well to those calculated using creel methods for non-lodge anglers (Figure 10; adjusted $r^2 = 0.88$) and for which the relationship slope was not significantly different from the 1:1 line ($t_{13} = 2.0$, $P = 0.07$). Estimated suggested that an additional 8-11% of fishing effort is occurring in April, that is non accounted for by conducting a creel survey from May to August. This amounts to fishing trips numbering from 766 to 1,303 depending on the year.

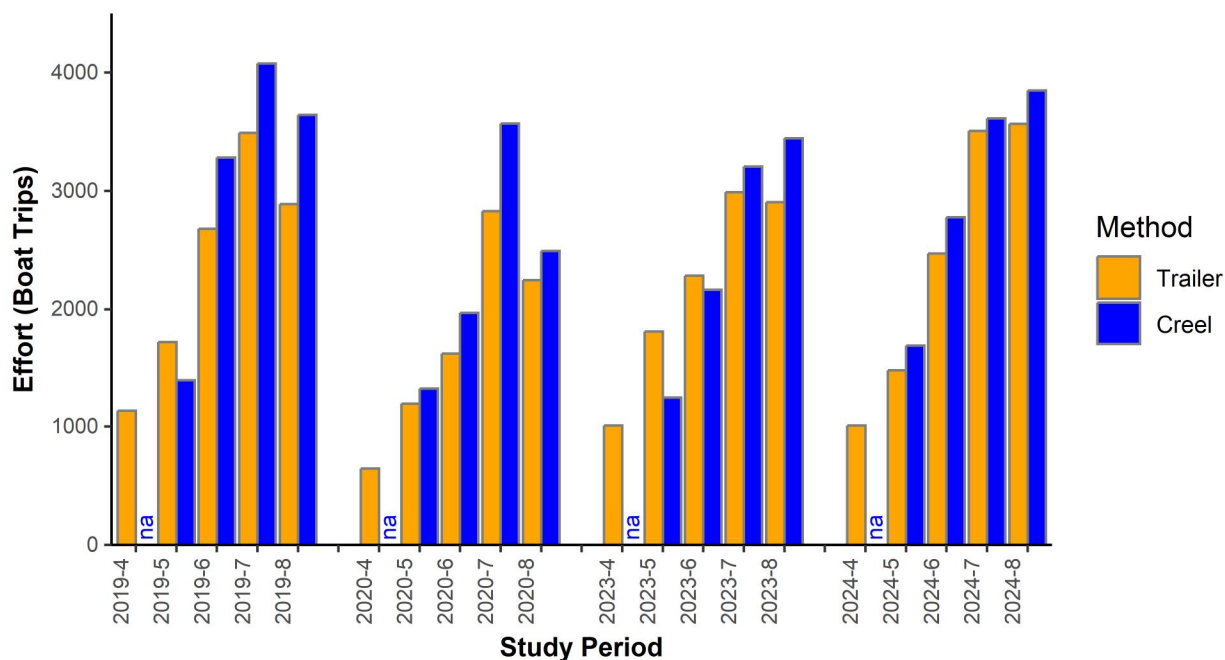


Figure 9. Total monthly non-lodge fishing effort as estimated using the usual creel methods (blue) or from trailer counts (orange; estimated using a combination of trailers counted at landing sites, the year-specific trailer-to-boat count regression, and the year-, month- and day type-specific activity pattern data). Creel-based estimates are not available for April (thus 'na').

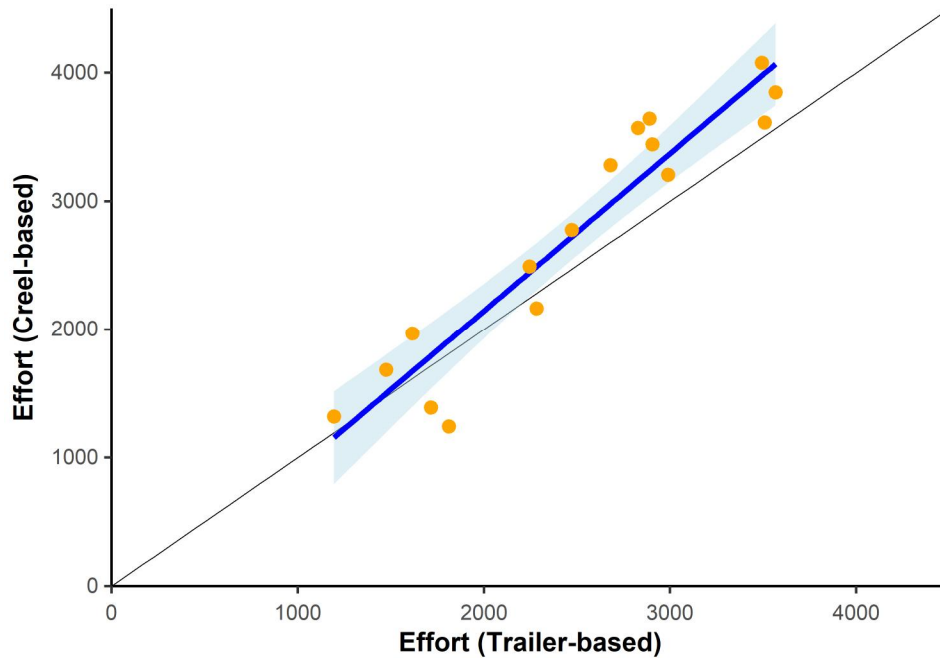


Figure 10. Relationship between total monthly non-lodge fishing effort values as estimated using the usual creel methods (Y axis) or from trailer counts (X axis). Blue line shows the line of best fit from a linear regression, and the thin black diagonal shows the 1:1 line.

Table 13. Estimated April non-lodge fishing effort in the Areas 3 and 4 in 2019, 2020, 2023, and 2024, along with data used in calculation steps. A: The index of effort for April as calculated using trailer count data, the count-to-boat regression, and angler activity profile data. B: The trailer count effort indices for May to August, summed. A/B: the ratio of A to B, used as a scaler. C: the creel-derived published estimates of non-lodge effort, summed over May to August. A/B(C): Estimates of April non-lodge fishing effort, calculated by applying the index scale to the creel-derived effort estimates from the May to August study period.

	A	B	A/B	C	A/B(C)
Year	April Index	May-Aug Index	Ratio of April to May-Aug Indices	May-Aug Non-lodge Creel Estimate	April Non-lodge Effort (scaled)
2019	1,133	10,777	11%	12,397	1,303
2020	646	7,884	8%	9,353	766
2023	1,010	9,989	10%	10,062	1,017
2024	1,009	11,022	9%	11,928	1,092

Catch Estimates

Monthly and full-season estimates were made of the numbers of fish released and retained, by species (Tables 14-23; Appendix 4). Non-lodge estimates were generated from the product of effort and CPUE estimates. Lodge catches were extracted from census documents that were submitted to DFO by the lodges, and which were then presented by DFO to NCSFNSS with identifying information removed for Privacy Act reasons. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 9,239 (SE = 223) Chinook Salmon were retained by anglers in Areas 3 or 4 during the 2024 study period (Table 15; Figure 11). The largest proportional harvest came from subareas 4F (35%), 4C (15%), 3J (14%), and 3K (11%). Harvest was highest in June (31% of total harvest), but the among-month differences were not statistically significant ($\chi^2 = 1.5$, $df = 3$, $P = 0.69$; Table 24). The number of Chinook Salmon released during the study was estimated at 10,605 (SE = 389), or 53% of the total catch. Percent retention (calculated for non-lodge anglers) was 64% in May, 40% in June, 29% in July, and 64% in August (Appendix 4).

The salmonid that was harvested in the greatest numbers during the 2024 study period was, by far, Coho Salmon (29,253 fish retained, SE = 530; Table 16; Figure 12). The largest proportional harvest came from subarea 4F (37%), followed by subareas 3J (18%), 3K (11%), and 4C (11%). Harvest differed significantly among months: harvest in May was significantly lower than all other months ($\chi^2 = 20.2$, $df = 3$, $P = 0.0002$; Table 24). During the study period, an estimated 3,011 (SE = 236) Coho Salmon were released by anglers (9% of the total catch).

Of the main three salmon species caught in 2024, Pink Salmon was the least likely to be retained (Table 17). We estimate that 16,170 (SE = 599) fish were caught, of which 13% were retained (2,036 fish, SE = 109), and 87% were released (14,134 fish, SE = 589). Harvest of Pink Salmon varied significantly among months ($\chi^2 = 18.8$, $df = 3$, $P = 0.0003$; Table 24): harvest in May was significantly lower than all other month, and that in June was significantly lower than in July or August.

Very few Chum (*Oncorhynchus keta*, 27 fish, SE = 9) or Sockeye (*O. nerka*, 45 fish, SE = 10) salmon were harvested in Area 3 and 4 during the study period (Tables 18 and 19). In 2024, a total of 58 harvested fish (SE = 24) were recorded as unknown salmon species.

Table 14. Total (lodge + non-lodge) estimated harvest, by subarea, for popular sportfish taxa, 2024.

Species	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
Chinook Salmon	554	1,319	1,055	5	351	1,340	709	6	3,248	266	231	154	2,928	6,311	9,239	138	174	223
Coho Salmon	2,626	5,392	3,329	0	327	3,157	1,849	17	10,844	920	693	100	11,346	17,907	29,253	377	373	530
Pink Salmon	176	255	115	0	51	410	160	0	706	74	89	0	546	1,490	2,036	73	82	109
Chum Salmon	0	3	9	0	4	6	0	0	5	0	0	0	12	15	27	7	5	9
Sockeye Salmon	0	4	8	0	0	16	2	0	16	0	0	0	11	34	45	6	8	10
Unknown Salmon	0	0	0	0	0	0	2	0	43	0	13	0	0	58	58	0	24	24
Pacific Halibut	1,072	3,700	716	1	325	1,651	748	461	2,982	1,442	1,358	500	5,488	9,469	14,957	189	242	308
Lingcod	263	4,201	107	0	19	324	231	349	1,722	764	437	316	4,571	4,162	8,733	232	222	322
Other Groundfish *	545	277	147	0	82	278	85	40	347	443	418	85	968	1,779	2,747	163	169	235
Rockfish †	470	2,234	344	2	442	620	600	393	1,604	514	500	183	3,048	4,857	7,906	189	250	313

* includes Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and unknown groundfish

† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

Table 15. Total (lodge + non-lodge) estimated Chinook Salmon catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	113.2	42.9	243.9	4.5	230.8	575.0	204.3	0	998.5	14.6	56.4	110.2	400	2,194	2,594	38	108	114
Released	41.9	13.8	91.9	3.6	200.0	344.3	68.8	0	578.1	14.6	0	0	148	1,209	1,357	29	91	95
June																		
Retained	164.9	413.9	343.5	0.4	92.1	247.9	263.7	0	1114.7	118.4	51.9	36.9	922	1,926	2,848	70	92	115
Released	154.8	531.1	1313.4	0.4	168.8	434.5	218.2	0	929.9	473.7	58.4	19.8	1,999	2,304	4,303	233	204	310
July																		
Retained	179.7	279.0	234.1	0	4.7	98.0	141.1	0	285.3	0	42.4	7.2	693	579	1,272	59	35	68
Released	569.3	565.9	779.4	0.5	80.1	428.6	167.2	10.7	572.8	111.2	144.7	3.1	1,915	1,519	3,433	140	99	171
August																		
Retained	96.3	583.3	233.1	0	23.4	419.1	100.1	5.8	849.8	133.3	80.7	0	913	1,612	2,525	97	96	136
Released	41.3	274.3	373.6	0.3	9.4	402.1	70.9	0	201.2	77.7	60.5	0	689	822	1,511	106	75	130
TOTAL																		
Retained	554	1,319	1,055	5	351	1,340	709	6	3,248	266	231	154	2,928	6,311	9,239	138	174	223
Released	807	1,385	2,558	5	458	1,609	525	11	2,282	677	264	23	4,751	5,854	10,605	293	256	389
Catch	1,361	2,704	3,613	10	809	2,949	1,234	17	5,530	944	495	177	7,679	12,165	19,844	324	310	448

Table 16. Total (lodge + non-lodge) estimated Coho Salmon catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	3.8	0	9.3	0	0	3.7	0	0	10.5	0	0	0	13	14	27	4	6	7
Released	0	0	0	0	0	0	0	0	10.5	0	0	0	0	11	11	0	7	7
June																		
Retained	161.1	433.4	470.3	0	7.7	399.8	57.6	0	880.2	282.4	64.8	51.0	1,065	1,744	2,808	89	149	174
Released	0	82.9	217.1	0	3.8	26.7	3.0	0	49.7	18.2	1.6	0	300	103	403	121	24	123
July																		
Retained	1938.5	2605.1	2045.9	0	155.5	1261.3	1220.2	0	5720.9	282.0	379.1	34.9	6,590	9,054	15,643	273	248	369
Released	154.6	195.8	197.0	0	4.7	110.2	267.0	42.9	595.2	11.9	2.5	0	547	1,034	1,582	116	122	168
August																		
Retained	522.2	2353.3	803.2	0	164.0	1492.4	571.5	17.4	4232.7	355.3	248.9	13.7	3,679	7,096	10,775	244	234	338
Released	28.7	174.2	37.1	0	16.4	184.0	150.2	46.3	333.8	44.4	0	0	240	775	1,015	78	77	110
TOTAL																		
Retained	2,626	5,392	3,329	0	327	3,157	1,849	17	10,844	920	693	100	11,346	17,907	29,253	377	373	530
Released	183	453	451	0	25	321	420	89	989	75	4	0	1,087	1,923	3,011	185	146	236
Catch	2,809	5,845	3,780	0	352	3,478	2,270	107	11,833	994	697	100	12,434	19,830	32,264	420	400	580

Table 17. Total (lodge + non-lodge) estimated Pink Salmon catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	3.7	0	0	0	0	0	0	0	4	4	0	3	3
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	20.4	15.6	0	0	3.8	0	6.1	0	52.4	9.1	0	0	36	71	107	13	14	19
Released	0	41.5	0	0	0	2.7	6.1	0	16.6	0	0	0	41	25	67	20	7	21
July																		
Retained	135.1	116.8	56.6	0	21.2	232.7	91.2	0	345.9	31.8	62.4	0	308	785	1,094	47	58	74
Released	481.6	1189.3	1262.4	0	108.3	921.5	744.7	0	3263.6	99.3	167.1	5.1	2,933	5,310	8,243	377	255	455
August																		
Retained	20.2	122.4	58.7	0	25.8	173.8	62.6	0	307.7	33.3	26.9	0	201	630	831	54	56	78
Released	661.1	779.6	222.7	0	72.6	923.4	450.5	17.4	2322.3	199.9	174.9	0	1,663	4,161	5,824	289	237	374
TOTAL																		
Retained	176	255	115	0	51	410	160	0	706	74	89	0	546	1,490	2,036	73	82	109
Released	1,143	2,010	1,485	0	181	1,848	1,201	17	5,602	299	342	5	4,638	9,496	14,134	476	348	589
Catch	1,318	2,265	1,600	0	232	2,258	1,361	17	6,309	373	431	5	5,184	10,986	16,170	481	358	599

Table 18. Total (lodge + non-lodge) estimated Chum Salmon catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	3.0	0	0	1.9	0	0	0	0	0	0	0	3	2	5	2	1	3
Released	25.8	8.9	0	0	0	5.3	0	0	55.2	36.4	8.1	0	35	105	140	14	34	37
July																		
Retained	0	0	9.0	0	0	6.1	0	0	0	0	0	0	9	6	15	7	4	8
Released	4.4	0	17.9	0	0	12.2	0	0	27.0	0	0	0	22	39	62	13	10	17
August																		
Retained	0	0	0	0	2.3	0	0	0	4.7	0	0	0	0	7	7	0	3	3
Released	0	24.9	0	0	0	13.6	2.1	0	7.1	11.1	0	0	25	34	59	13	11	17
TOTAL																		
Retained	0	3	9	0	4	6	0	0	5	0	0	0	12	15	27	7	5	9
Released	30	34	18	0	0	31	2	0	89	48	8	0	82	178	260	24	37	44
Catch	30	37	27	0	4	37	2	0	94	48	8	0	94	193	287	25	37	45

Table 19. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
June																		
Retained	0	0	7.8	0	0	2.7	0	0	0	0	0	0	8	3	10	6	2	6
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
July																		
Retained	0	3.7	0	0	0	6.1	0	0	13.5	0	0	0	4	20	23	3	7	7
Released	0	0	0	0	0	0	6.5	0	6.7	0	0	0	0	13	13	0	6	6
August																		
Retained	0	0	0	0	0	6.8	2.1	0	2.4	0	0	0	0	11	11	0	4	4
Released	9.6	8.3	0	0	0	0	2.1	0	0	0	0	0	18	2	20	9	2	9
TOTAL																		
Retained	0	4	8	0	0	16	2	0	16	0	0	0	11	34	45	6	8	10
Released	10	8	0	0	0	0	9	0	7	0	0	0	18	15	33	9	6	11
Catch	10	12	8	0	0	16	11	0	23	0	0	0	29	49	78	11	10	15

Table 20. Total (lodge + non-lodge) estimated Pacific Halibut catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	138.7	286.7	79.0	0	61.5	175.8	112.6	125.8	353.9	197.4	143.3	120.2	504	1,291	1,795	45	91	101
Released	149.0	153.2	125.1	0	7.7	161.2	35.4	47.2	308.3	241.3	86.8	76.8	427	965	1,392	75	139	158
June																		
Retained	106.7	641.6	233.2	0.4	59.5	261.2	194.0	100.9	731.2	692.4	118.3	235.4	982	2,393	3,375	61	140	153
Released	0	319.9	62.0	0	19.2	215.9	66.7	100.9	491.1	892.8	63.2	127.6	382	1,977	2,359	65	268	275
July																		
Retained	303.5	1064.7	277.4	0.5	117.8	471.4	254.0	60.7	1089.4	286.0	376.6	83.1	1,646	2,740	4,385	90	102	136
Released	167.9	856.9	143.3	0	108.3	275.5	86.8	200.0	680.6	127.1	503.8	84.2	1,168	2,066	3,234	183	183	259
August																		
Retained	522.9	1706.8	126.6	0	86.7	742.8	187.7	173.7	807.2	266.5	719.7	61.6	2,356	3,046	5,402	148	143	206
Released	450.3	1235.7	0	0	23.4	310.1	89.7	57.9	466.3	122.2	578.5	44.5	1,686	1,693	3,379	248	195	315
TOTAL																		
Retained	1,072	3,700	716	1	325	1,651	748	461	2,982	1,442	1,358	500	5,488	9,469	14,957	189	242	308
Released	767	2,566	330	0	159	963	279	406	1,946	1,383	1,232	333	3,663	6,701	10,364	324	403	517
Catch	1,839	6,266	1,047	1	484	2,614	1,027	867	4,928	2,826	2,590	833	9,151	16,170	25,321	375	470	601

Table 21. Total (lodge + non-lodge) estimated Lingcod catch, by subarea, shown separately for released and retained fish by month, 2024.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	168.0	492.1	13.1	0	3.8	73.3	68.8	149.4	266.3	43.9	78.1	120.2	673	804	1,477	87	124	151
Released	40.8	121.3	0	0	7.7	25.6	4.2	0	17.5	0	21.7	0	162	77	239	60	20	63
June																		
Retained	0	749.9	52.1	0	3.8	32.0	66.7	56.0	416.6	337.1	74.6	127.6	802	1,114	1,916	58	116	129
Released	0	23.7	31.0	0	24.9	0	9.1	16.8	52.4	9.1	11.3	0	55	124	178	18	27	33
July																		
Retained	83.1	1638.9	42.1	0	9.4	116.3	43.4	85.7	712.0	305.8	102.3	26.7	1,764	1,402	3,166	117	106	158
Released	0	81.3	0	0	4.7	15.3	10.9	7.1	42.7	15.9	5.0	0	81	102	183	24	16	29
August																		
Retained	12.0	1319.8	0	0	2.3	102.2	52.1	57.9	326.7	77.7	181.6	41.1	1,332	842	2,174	171	98	197
Released	0	58.1	0	0	4.7	3.4	4.2	0	40.2	33.3	33.6	0	58	119	178	32	35	48
TOTAL																		
Retained	263	4,201	107	0	19	324	231	349	1,722	764	437	316	4,571	4,162	8,733	232	222	322
Released	41	284	31	0	42	44	28	24	153	58	72	0	356	421	778	74	52	90
Catch	304	4,485	138	0	61	368	259	373	1,874	823	508	316	4,927	4,583	9,510	244	228	334

Table 22. Total (lodge + non-lodge) estimated catch of other groundfish by subarea, shown separately for released and retained fish by month, 2024. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	40.8	12.8	20.0	0	19.2	36.6	14.6	0	171.7	51.2	39.1	56.8	74	389	463	23	70	74
Released	8.2	19.1	5.0	0	107.7	25.6	18.8	102.2	49.1	21.9	21.7	0	32	347	379	15	99	100
June																		
Retained	32.3	3.0	15.5	0	23.0	50.6	30.3	11.2	46.9	227.8	42.1	19.8	51	452	503	19	95	97
Released	12.9	5.9	15.5	0	26.9	45.3	45.5	5.6	91.1	45.6	19.5	8.5	34	288	322	11	59	60
July																		
Retained	145.8	70.2	0	0	23.6	58.2	8.7	0	76.4	19.9	94.8	0	216	281	497	48	34	59
Released	48.6	33.2	0	0	101.3	159.2	54.3	0	60.6	11.9	82.3	0	82	470	551	27	63	69
August																		
Retained	325.7	190.7	111.3	0	16.4	132.9	31.3	29.0	52.1	144.4	242.2	8.6	628	657	1,285	153	117	193
Released	143.7	124.4	37.1	0.6	51.5	371.4	33.4	0	127.8	55.5	470.8	0	305	1,111	1,416	68	231	241
TOTAL																		
Retained	545	277	147	0	82	278	85	40	347	443	418	85	968	1,779	2,747	163	169	235
Released	213	183	58	1	287	602	152	108	329	135	594	9	454	2,216	2,669	75	266	276
Catch	758	459	204	1	370	880	237	148	676	578	1,012	94	1,422	3,995	5,416	180	315	363

Table 23. Total (lodge + non-lodge) estimated rockfish catch, by subarea, shown separately for released and retained fish by month, 2024. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, and Yellowtail rockfish, and other miscellaneous rockfishes.

Month / Catch Type	Subarea												TOTALS			SE of the Totals		
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
May																		
Retained	137.1	248.9	141.1	2.2	200.0	44.0	120.9	39.3	245.3	14.6	95.5	93.5	527	855	1,382	62	102	119
Released	63.3	140.4	40.0	1.3	115.4	183.1	156.4	31.5	231.2	87.7	21.7	26.7	244	855	1,099	72	97	121
June																		
Retained	142.3	200.6	100.8	0	65.2	135.9	218.2	207.3	281.4	291.5	50.2	53.9	444	1,304	1,747	60	181	191
Released	64.5	234.0	23.3	0.7	149.6	119.9	309.1	78.4	358.7	610.4	13.0	14.2	322	1,654	1,976	79	231	245
July																		
Retained	132.9	726.8	0	0.3	108.3	150.0	156.3	100.0	781.7	174.8	112.2	21.6	860	1,605	2,465	76	99	125
Released	234.2	598.3	62.7	0	171.9	254.1	191.1	32.2	833.3	91.4	84.8	16.4	895	1,675	2,570	117	119	167
August																		
Retained	57.5	1058.1	102.4	0	68.0	289.6	104.3	46.3	295.9	33.3	242.2	13.7	1,218	1,093	2,311	149	98	178
Released	124.6	431.3	111.3	0	107.8	231.7	216.9	11.6	359.8	344.2	114.3	15.4	667	1,402	2,069	116	194	226
TOTAL																		
Retained	470	2,234	344	2	442	620	600	393	1,604	514	500	183	3,048	4,857	7,906	189	250	313
Released	486	1,404	237	2	545	789	873	154	1,783	1,134	234	73	2,128	5,586	7,714	196	339	392
Catch	956	3,638	582	5	986	1,408	1,473	547	3,387	1,648	734	255	5,176	10,444	15,620	272	421	502

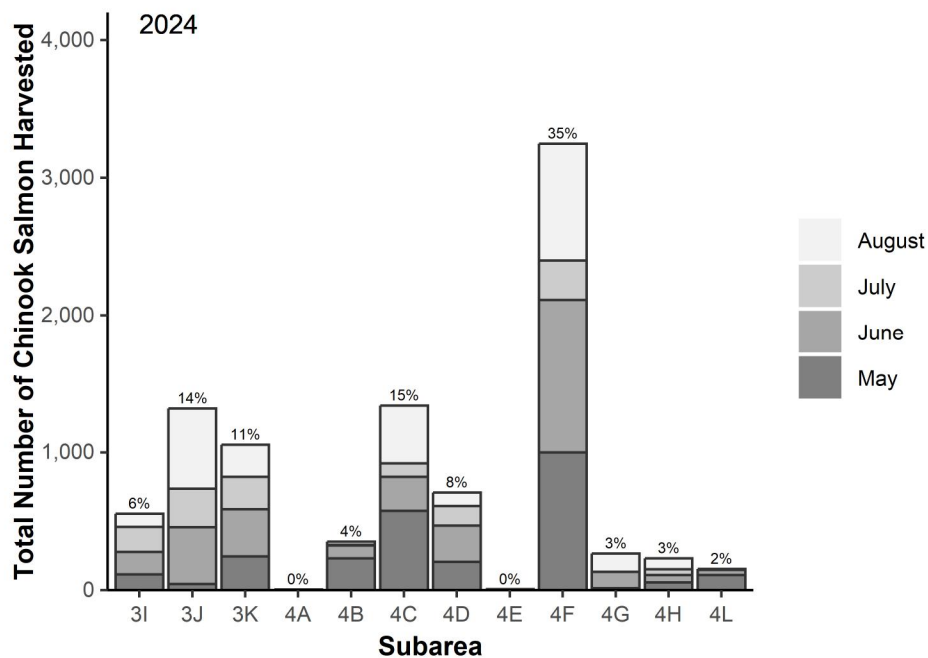


Figure 11. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2024.

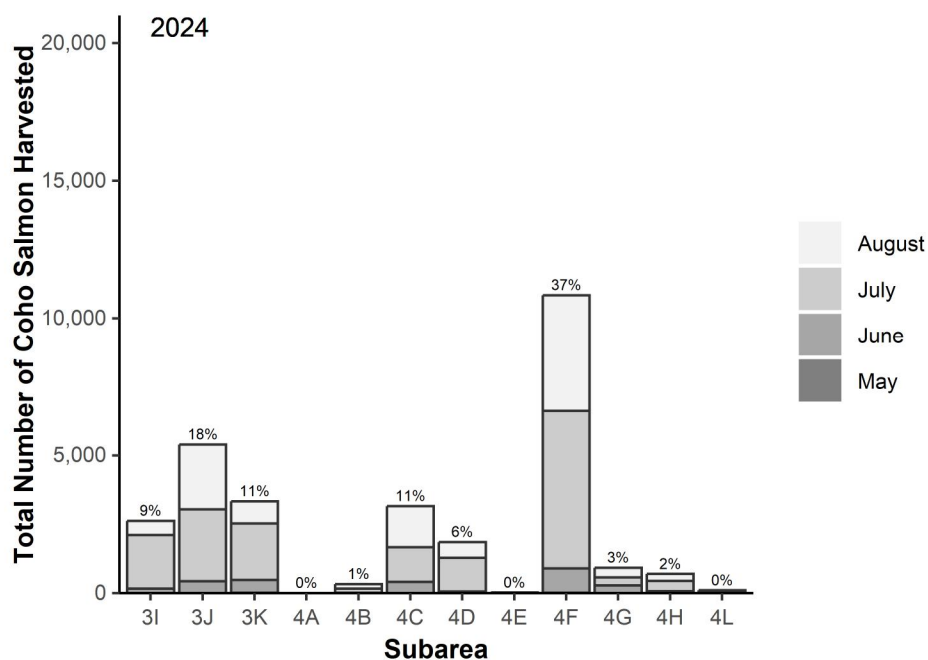


Figure 12. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2024.

Table 24. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2024. Subareas treated as replicates. Red letters indicate statistical differences within species (months not connected by a common letter were statistically different).

Species	Kruskal-Wallis Statistics			Median Monthly Harvest			
	χ^2	df	P	May	June	July	August
<i>Chinook Salmon</i>	1.5	3	0.69	88.8	107.6	70.2	90.4
<i>Chum Salmon</i>	2.3	3	0.52	0	0	0	0
<i>Coho Salmon</i>	20.2	3	0.0002	0 B	77.6 A	799.6 A	340.5 A
<i>Pink Salmon</i>	18.8	3	0.0003	0 C	1.9 B	58.0 A	30.1 A
<i>Sockeye Salmon</i>	3.5	3	0.32	0	0	0	0
<i>Arrow-tooth Flounder</i>	6.7	3	0.08	0	0	0	0
<i>Big Skate</i>	2.0	3	0.56	0	0	0	0
<i>Cabazon</i>	4.7	3	0.19	0	0	0	0
<i>English Sole</i>	2.0	3	0.58	0	0	0	0
<i>Greenlings</i>	2.2	3	0.53	2.2	0	0	0
<i>Lingcod</i>	0.6	3	0.91	75.7	61.4	69.4	55.0
<i>Longnose Skate</i>	3.0	3	0.39	0	0	0	0
<i>Pacific Cod</i>	4.4	3	0.22	13.9	21.4	9.9	74.6
<i>Pacific Halibut</i>	3.3	3	0.34	123.0	174.5	239.7	227.1
<i>Pacific Rock Sole</i>	2.6	3	0.46	0	0	0	0
<i>Pacific Spiny Dogfish</i>	4.3	3	0.23	0	0	0	0
<i>Sablefish</i>	3.0	3	0.39	0	0	0	0
<i>Spotted Ratfish</i>	0.0	3	1	0	0	0	0
<i>Starry Flounder</i>	2.0	3	0.58	0	0	0	0
<i>Black Rockfish</i>	1.3	3	0.73	19.1	30.9	34.5	21.4
<i>Boccacio Rockfish</i>	1.0	3	0.81	0	0	0	0
<i>Canary Rockfish</i>	3.1	3	0.38	0	0	0	0
<i>China Rockfish</i>	1.8	3	0.61	4.7	0	1.1	0
<i>Copper Rockfish</i>	1.2	3	0.75	6.3	0	1.7	0
<i>Dusky Rockfish</i>	0.7	3	0.88	0	0	0	0
<i>Quillback Rockfish</i>	0.9	3	0.83	13.1	12.3	32.7	22.8
<i>Redbanded Rockfish</i>	0.0	3	1	0	0	0	0
<i>Redstripe Rockfish</i>	6.1	3	0.11	0	0	0	0
<i>Tiger Rockfish</i>	8.9	3	0.030	0 A	0 A	3.5 A	1.0 A
<i>Widow Rockfish</i>	2.2	3	0.53	0	0	0	0
<i>Yelloweye Rockfish</i>	2.2	3	0.53	0	0	0	0
<i>Yellowtail Rockfish</i>	2.0	3	0.58	0	4.5	2.5	0

An estimated 14,957 (SE = 308) Pacific Halibut were harvested in Areas 3 or 4 during the 2024 study period (Table 20; Figure 13). Most of the harvest was attributed to subareas 3J (25%), 4F (20%), and 4C (11%). Harvest did not vary significantly among months ($\chi^2 = 3.3$, df = 3, $P = 0.34$; Table 24). In all, 10,364 (SE = 517) Pacific Halibut were released, representing 41% of the total catch.

From May to August 2024, an estimated 8,733 (SE = 322) Lingcod were harvested, occurring mainly from subareas 3J (48%) and 4F (20%; Table 21; Figure 14). Harvest did not vary significantly among months ($\chi^2 = 0.6$, df = 3, $P = 0.91$; Table 24). Of the captured Lingcod, 8% were released (778 fish, SE = 90).

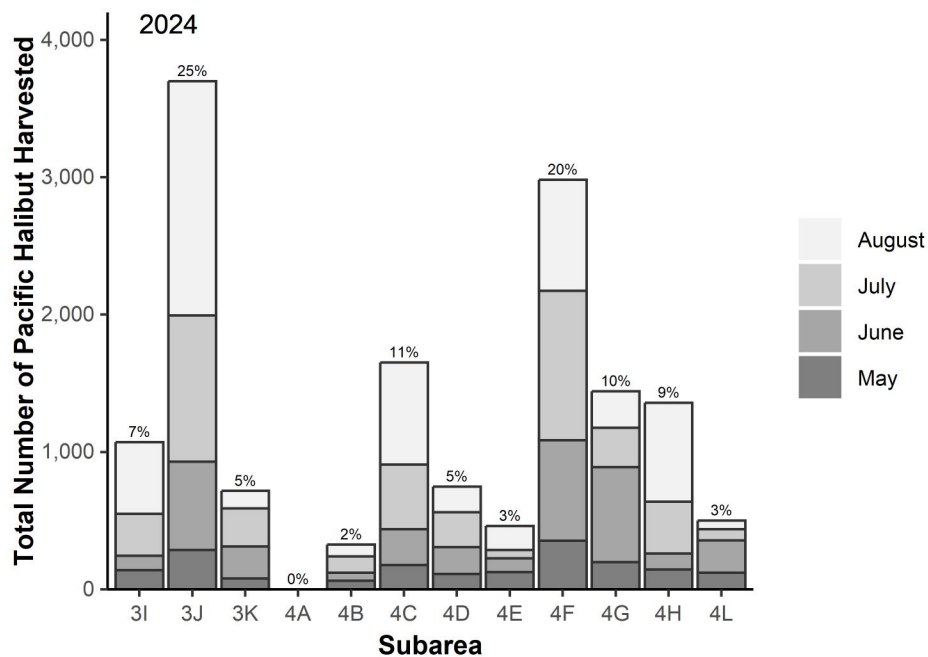


Figure 13. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2024.

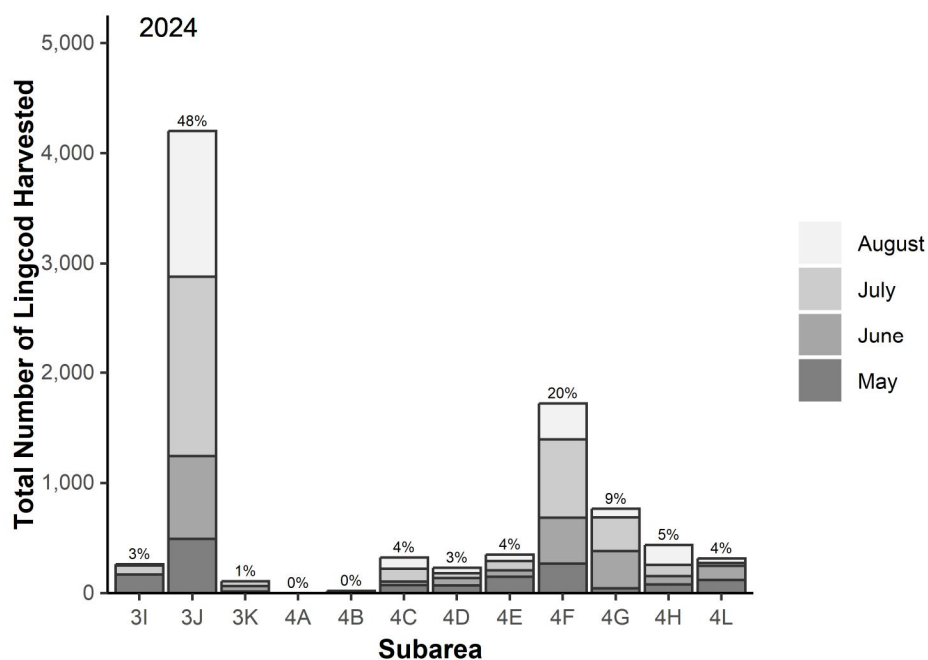


Figure 14. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2024.

An estimated 2,747 (SE = 235) fish from other groundfish species were harvested from May to August 2024, of which 86% (i.e., 2,365 fish, SE = 230) were Pacific Cod. The remaining harvest included greenlings (*Hexagrammos* spp.; 6%), Pacific Rock Sole (*Lepidopsetta bilineata*, 2%), Cabezon (*Scorpaenichthys marmoratus*, 1%), and a variety of other species (<35 individuals retained per species). Over the study period, it is estimated that 458 (SE = 69) Pacific Cod were released (16% of the total catch of Pacific Cod).

Rockfish harvest was estimated at 7,906 fish (SE = 313; Table 23), composed mostly of Black (35%), Quillback (24%), Yellowtail (*Sebastes flavidus*, 5%), Copper (*S. caurinus*, 4%), China (*S. nebulosus*, 3%), Tiger (3%), Bocaccio (*S. paucispinis*, 2%), and unknown (23%) rockfish, along with six other rockfish species. In all, 7,714 (SE = 392) were released. Of the 1,409 Yelloweye Rockfish caught, all but 53 (96%) were released.

Table 25. Biosampling results, 2024.

	Chinook Salmon		Coho Salmon		Pacific Halibut	
	Number	Percent	Number	Percent	Number	Percent
Number Examined	483		725		391	
Subarea						
3I	3	1%	18	2%	4	1%
3J	28	6%	38	5%	59	15%
3K	12	2%	17	2%	2	1%
4A	3	1%				
4B	60	12%	32	4%	46	12%
4C	101	21%	140	19%	55	14%
4D	73	15%	100	14%	37	10%
4E					11	3%
4F	193	40%	355	49%	106	27%
4G	1	0%	16	2%	25	6%
4H	9	2%	9	1%	38	10%
4L					4	1%
Sex						
Male	4	50%	1	17%		
Female	4	50%	5	83%		
Undeterminable	5		4			
Unknown	470		715		391	
Trip Type						
Guided	199	41%	296	41%	160	41%
Unguided	284	59%	429	59%	231	59%
Adipose Clips						
Clipped	98	21%	5	1%		
Unmarked	377	79%	707	99%		
Unknown						
Head Tagged						
Yes	47	10%	3	0.4%		
No	436	90%	722	100%		
Scale Samples			Whatman Samples			
Yes	480	99%	725	100%		
No	3	1%				
Flesh Colour						
Red	51	98%				
White	1	2%				
Marble						
Unknown						

Biosampling

During interviews, observers were told about 2,494 retained Chinook Salmon, and were permitted to handle 483 fish. Too few fish were sexed ($n=8$) to provide a robust estimate of sex ratio (Table 25). The median fork length of biosampled Chinook Salmon ($n = 479$) was 69 cm (95% of observations were between 52 and 91 cm; Figure 15). In all, 98 adipose-clipped individuals were encountered (Table 25), including 39 from standardized sampling methods (complete samples collected from anglers who were asked if their catch could be biosampled before any other interactions took place; Appendix 5) and 60 from non-standard interactions (“non-standard interactions” were interviews for which anything other than complete access was given to all harvested fish). In all, 47 Chinook Salmon had their heads collected and labelled for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples ($n=210$), 18.6% were adipose-clipped (Appendix 5). Scale samples were obtained from 99% ($n=480$) of the biosampled Chinook Salmon (Table 25).

For Coho Salmon, observers were told about 8,387 retained fish, and were permitted to handle 725 fish. Too few fish were sexed ($n=6$) to provide a robust estimate of sex ratio (Table 25). The median Coho Salmon fork length ($n = 724$) was 62 cm (95% of observations were between 50 and 71 cm; Figure 15). In all, 5 adipose-clipped individuals were encountered (Table 25), none from standardized sampling interviews. Three heads were collected for submission to the Salmon Head Recovery Program (Table 25). Of the standardized samples ($n=228$), 0% were adipose-clipped (Appendix 5). Tissue samples were obtained from all ($n=725$) of the biosampled Coho Salmon (Table 25).

Of the 3,844 Pacific Halibut that anglers reported as retained, 391 were handled (Table 25). All were measured, but none were sexed. The median length was 80 cm (95% of observations were between 59 and 124 cm; Figure 15). Average net and round weights, calculated from average lengths using regression formulas, were 7.6 kg (16.8 lb) and 10.1 kg (22.2 lb) in Area 3, and 5.6 kg (12.4 lb) and 7.5 kg (16.4 lb) in Area 4, respectively (Table 26).

Table 26. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2024.

	Stat Area 3				Stat Area 4			
	Net Weight		Round Weight		Net Weight		Round Weight	
	kg	lb	kg	lb	kg	lb	kg	lb
Creel Interview (Non-Lodge) Average Weights								
Guided	8.0	17.7	10.7	23.5	6.8	14.9	9.0	19.8
Unguided	5.9	13.0	7.8	17.2	5.1	11.2	6.7	14.8
Overall	7.6	16.8	10.1	22.2	5.6	12.4	7.5	16.4
Estimated Non-Lodge Harvest (number of fish)								
Overall		4,362				9,469		
Estimated Lodge Harvest (number of fish)								
Overall		1,126				0		
Biomass Harvested								
Non-lodge	33,239	73,280	44,021	97,050	53,338	117,589	70,639	155,733
Lodge	8,580	18,916	11,363	25,052	0	0	0	0
Overall	41,819	92,196	55,384	122,102	53,338	117,589	70,639	155,733

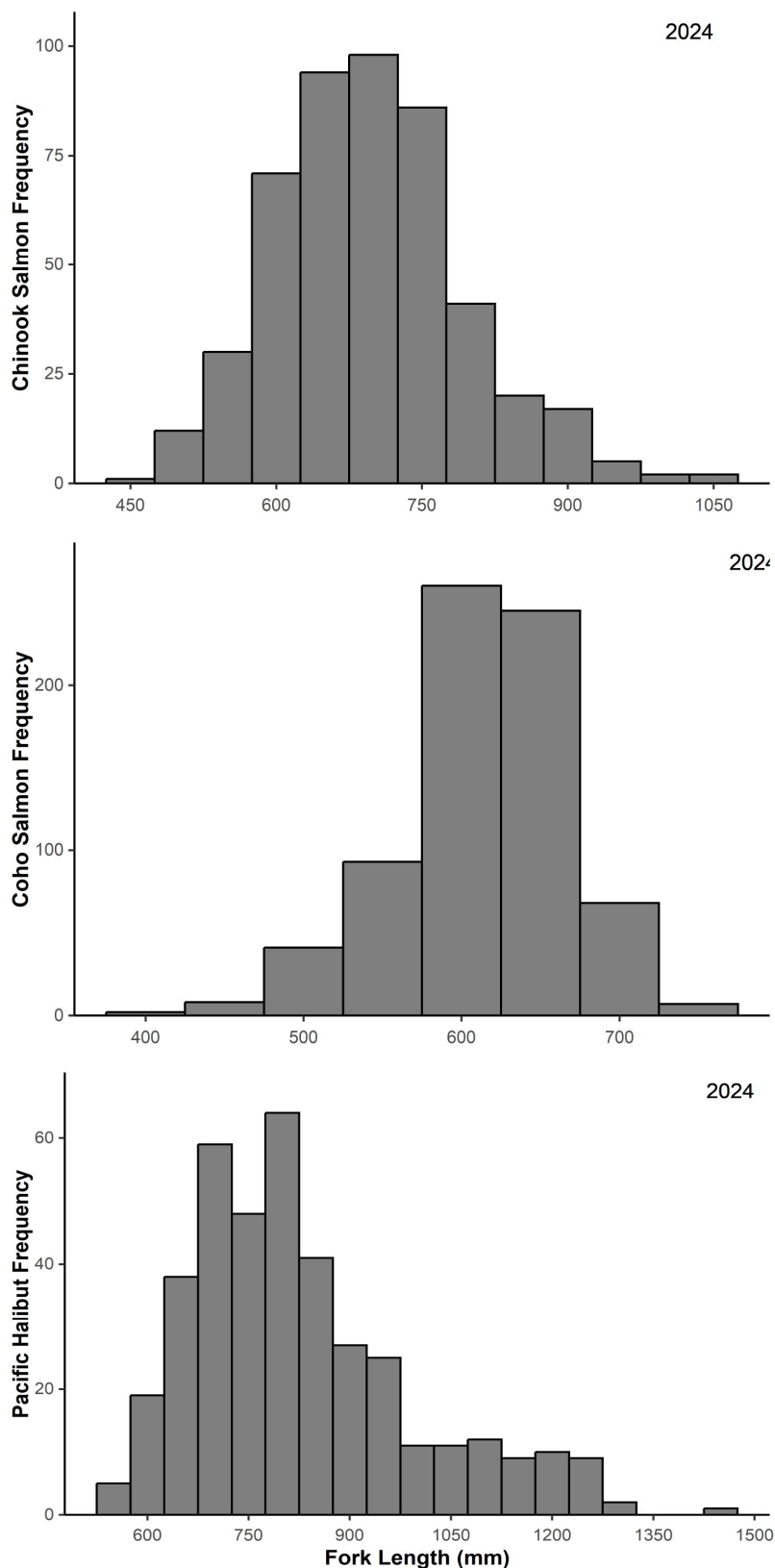


Figure 15. Length-frequency distributions of biosampled Chinook ($n = 479$) and Coho ($n = 724$) salmon, and Pacific Halibut ($n = 391$), 2024. Note that X and Y axis scales vary among plots.

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 95.2 tonnes (209,785 lb), including 41.8 t (92,196 lb) in Area 3 and 53.3 t (117,589 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 126.0 t (277,834 lb) including 55.4 t (122,102 lb) in Area 3 and 70.6 t (155,733 lb) in Area 4 (Table 26).

DISCUSSION

Creel Estimates

In general, recreational fishing effort and salmon catch in Areas 3 & 4 have been increasing over the last 30 years (Tables 27 and 28). Using published creel data collected since 1995, we found 16 years (1995-96, 2001-02, 2008-09¹, and 2015-2024) for which effort data were collected in June, July, and August (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017, Robichaud and Addison 2018, 2019, 2020, 2021, 2022, 2023, 2024), and we have therefore limited this meta analyses to those three months. Meta-analysis revealed (Figure 16) that fishing effort has increased significantly (by 180.7 trips per year on average; $F_{1,13} = 17.5$, $P = 0.0009$) since 1995. Effort in 2024 was close to the expected value given the long-term trend, whereas effort from 2020-2023 has been lower than what would have been predicted.

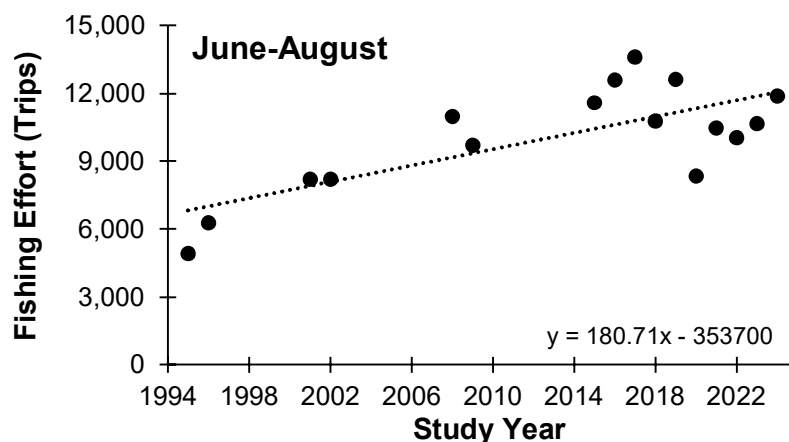


Figure 16. Fishing effort from 1995 to 2024. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Long-term linear regression equation is shown in bottom right of the panel.

¹ Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

Over the same 30-year period, Coho Salmon harvests have increased significantly by 862 fish per year, on average ($F_{1,13} = 11.9$, $P = 0.004$, Table 28, Figure 17). The 2024 Coho Salmon harvest was below that expected given the long-term trend.

The number of Chinook Salmon being released by anglers has been increasing strongly since 2009 (the first year with data available; ln scale, $F_{1,9} = 26.6$, $P = 0.0006$; Figure 18). Chinook Salmon harvest had been on an increasing trend until 2019, but now appears to be declining over time. The long-term trend (56 additional fish per year on average) is not statistically significant ($F_{1,14} = 0.4$, $P = 0.55$), whereas the trend since 2015 (791 fewer fish per year on average) is strongly and significantly negative ($F_{1,8} = 8.6$, $P = 0.02$). The dome-shaped long term trend may be the result of management measures to control harvest (as evidenced by the increasing trend in Chinook salmon release numbers), or the result of declining stocks.

Pacific Halibut harvest did not show a strong among-year trend (Figure 17). While fishing effort increased over the 30-year period, the total harvest of Pacific Halibut in recent years (2009, 2015-2024) has been very similar to that from 2001-2002; Table 28). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 17).

The Lingcod harvest (June to August) for the last four years (2021-2024) are the four highest observed for the available time series (Figure 17). In general, the harvest from 2018 to 2024 has been higher than in other recent years (e.g., 1,803 to 3,052 in 2015-2017). The long-term trend is statically significant (ln scale, $F_{1,13} = 7.5$, $P = 0.02$; Figure 17). This, despite a coast-wide reduction in the Lingcod bag limit from three to two fish starting in May 2018. The elevated and harvest values from 2021 and 2024 could potentially cause concern.

The meta-analysis results make three important points: 1) the effort and CPUE of certain species has increased tremendously over the past 30 years (even despite the COVID-19 pandemic), potentially imperilling stocks if left unchecked; 2) management decisions for Pacific Halibut and possibly Chinook Salmon appear to have been effective, indicating that tools are available to keep effort and CPUE increases from going unchecked; and 3) despite that, Lingcod harvest is increasing to a concerning degree.

May Effort

In our report for the 2018 study year (Robichaud and Addison 2019), we made the case that substantial angling activity was being missed by starting the creel survey in June each year. Surveys in 1995 and 1996 (Van Tongeren and Winther 2010) included May, which accounted for 14% and 5% of the May-August effort, and for 8% and 1% of the May-August Chinook harvest. May surveys occurred again in 2001, 2002, and 2008, representing 5%, 10%, and 8% of the May-August effort, respectively (and 5%, 7%, and 4% of the May-August Chinook harvest). Beginning in 2019, we did start the survey in May, and found that May effort (1,553 boat trips in 2019, 1,320 in 2020, 1,785 in 2021, 1,587 in 2022, 1,428 in 2023, and 1,851 in 2024) accounted for 11-15% of the May-August total. May effort from 2019 to 2021 and in 2023 accounted for 15-21% of the May-August total Chinook Salmon harvest, whereas effort in May 2022 and 2024 accounted for 32% and 28% of the Chinook Salmon harvest, respectively.

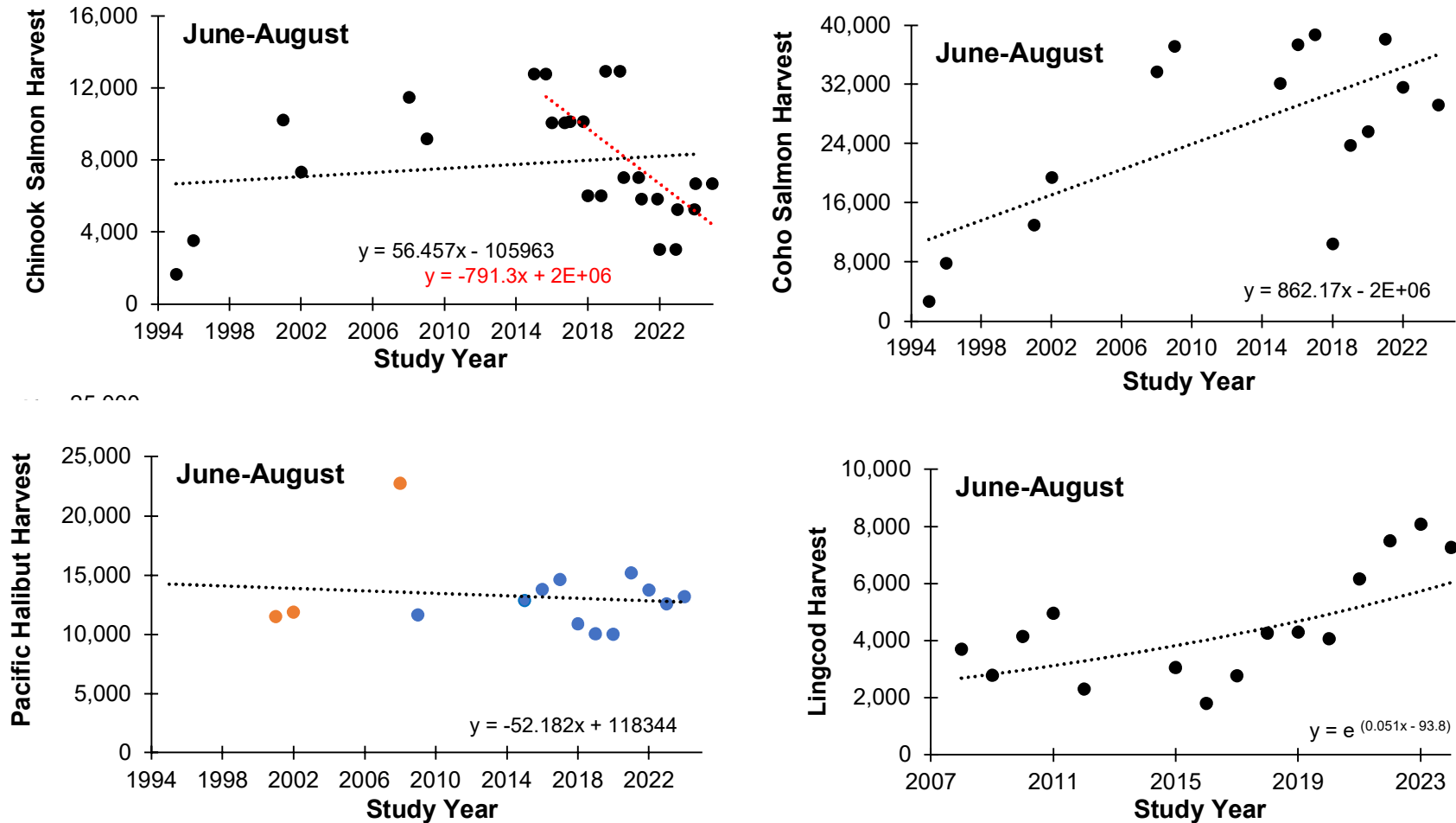


Figure 17. Harvest of Chinook and Coho salmon, Pacific Halibut, and Lingcod from 1995 to 2024. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Long-term linear regression equations are shown in black within each panel, and the Chinook Salmon harvest trend since 2015 is in red. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

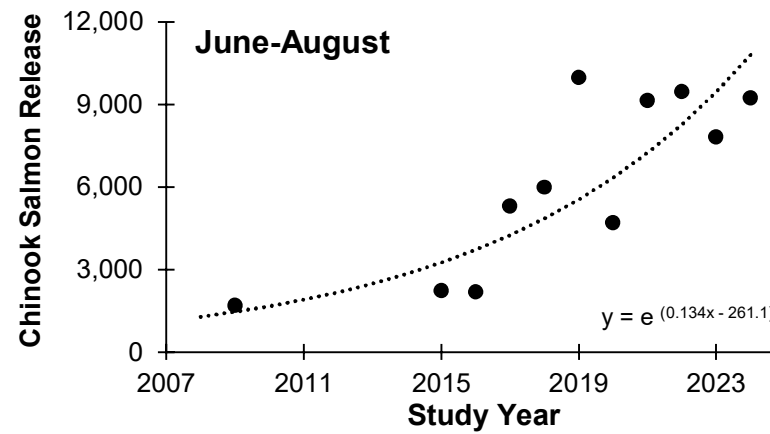


Figure 18. Number of Chinook Salmon released from June to August by anglers in Areas 3 and 4 from 1995 to 2024. Only years with data from all three months are included. Long-term linear regression equation (calculated on ln scale) is shown in the bottom right.

Table 27. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2024, as estimated using creel survey methods.

	Year																			
Month	1995	1996	1998	2000	2001	2002	2006	2008	2009		2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
May	965	391	--	--	449	960	--	983	--		--	--	--	--	1,553	1,320	1,785	1,587	1,428	1,851
June	1,590	2,490	--	--	3,801	3,463	--	4,696	3,043		3,424	3,961	4,574	2,873	3,790	1,970	2,507	2,712	2,735	3,335
July	1,394	2,385	1,307	1,842	2,588	2,892	--	3,750	3,804		4,279	5,295	6,044	4,593	4,644	3,745	4,318	4,040	3,868	4,247
Aug	1,927	1,400	1,351	1,327	1,808	1,843	2,593	2,529	2,865		3,881	3,325	2,980	3,304	4,175	2,631	3,642	3,285	4,055	4,296
Sept	864	488	504	954	540	676	758	394	817		--	--	--	--	--	--	--	--	--	--
TOTAL	6,740	7,154	3,162	4,123	9,186	9,834	3,351	12,352	10,529		11,585	12,581	13,598	10,770	14,163	9,665	12,252	11,624	12,086	13,729

Table 28. Monthly Chinook Salmon, Coho Salmon, Pacific Halibut, and Lingcod harvest, and Chinook Salmon releases, in the Area 3 and 4 recreational fishery from 1995 to 2024, as estimated using creel survey methods.

Month	Year																					
	1995	1996	1998	2000	2001	2002	2006	2008	2009	2010	2011	2012	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Chinook Salmon Harvest																						
May	141	45	--	--	509	522	--	474	--	--	--	--	--	--	--	--	2,234	1,256	1,580	1,428	1,191	2,594
June	907	2,165	--	--	6,630	4,028	--	7,708	4,797	--	--	--	7,071	5,678	5,229	1,273	5,675	2,574	1,834	1,564	3,180	2,848
July	505	1,097	254	1,611	3,260	2,822	--	3,080	3,873	--	--	--	4,016	3,474	3,800	2,964	4,806	2,856	2,797	9	1,155	1,272
Aug	245	257	35	108	299	474	1,207	688	503	--	--	--	1,669	891	1,078	1,593	2,437	1,560	1,172	1,431	917	2,525
Sept	15	62	6	93	61	59	203	20	4	--	--	--	--	--	--	--	--	--	--	--	--	--
TOTAL	1,813	3,626	295	1,812	10,759	7,905	1,410	11,970	9,177				12,756	10,043	10,107	5,830	15,152	8,247	7,383	4,432	6,443	9,239
Chinook Salmon Release																						
May	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	1,145	933	497	536	980	1,357
June	--	--	--	--	--	--	--	--	808	--	--	--	1,534	975	2,800	2,121	4,735	2,088	3,652	2,297	3,790	4,303
July	--	--	--	--	--	--	--	--	705	--	--	--	594	710	1,811	2,612	2,382	1,560	3,694	6,213	3,415	3,433
Aug	--	--	--	--	--	--	--	--	175	--	--	--	103	506	697	1,264	2,866	1,061	1,799	966	627	1,511
Sept	--	--	--	--	--	--	--	--	15	--	--	--	--	--	--	--	--	--	--	--	--	--
TOTAL									1,703				2,231	2,190	5,308	5,997	11,129	5,642	9,643	10,012	8,812	10,605
Coho Salmon Harvest																						
May	0	0	--	--	0	0	--	0	--	--	--	--	--	--	--	--	29	0	0	3	2	27
June	335	474	--	--	1,771	1,332	--	3,913	3,686	--	--	--	4,436	4,389	1,839	126	1,475	1,006	3,038	1,356	2,555	2,808
July	402	3,565	1,937	568	4,736	9,798	--	15,982	17,043	--	--	--	15,101	21,761	22,871	5,758	10,341	16,304	17,685	13,929	17,360	15,643
Aug	1,907	3,771	1,500	2,125	6,488	8,292	15,647	13,791	16,459	--	--	--	12,612	11,200	14,004	4,580	11,974	8,288	17,403	16,331	22,608	10,775
Sept	1,115	1,417	79	790	1,006	1,010	3,215	733	3,136	--	--	--	--	--	--	--	--	--	--	--	--	--
TOTAL	3,759	9,227	3,516	3,483	14,001	20,432	18,862	34,419	40,324				32,148	37,350	38,714	10,464	23,820	25,599	38,126	31,620	42,525	29,253
Pacific Halibut Harvest																						
May	--	--	--	--	281	467	--	1,435	--	--	--	--	--	--	--	--	941	867	1,499	1,385	934	1,795
June	--	--	--	--	4,872	3,912	--	8,367	3,029	--	--	--	3,551	4,032	4,632	3,407	2,698	2,344	3,725	4,229	2,636	3,375
July	--	--	--	3,963	3,535	4,716	--	8,507	4,466	--	--	--	4,096	6,069	6,563	4,263	3,894	4,150	6,518	5,558	4,887	4,385
Aug	--	--	1,801	2,530	3,085	3,247	8,178	5,865	4,142	--	--	--	5,201	3,670	3,410	3,183	3,435	3,504	4,929	3,953	5,049	5,402
Sept	--	--	328	1,224	222	615	1,812	774	1,205	--	--	--	--	--	--	--	--	--	--	--	--	--
TOTAL			2,129	7,717	11,995	12,957	9,990	24,948	12,842				12,848	13,771	14,605	10,853	10,967	10,865	16,671	15,125	13,505	14,957
Lingcod Harvest																						
May	--	--	--	--	--	--	--	81	--	--	--	--	--	--	--	--	219	635	1,178	1,661	802	1,477
June	--	--	--	--	--	--	--	1,462	760	1,093	2,241	692	1,042	732	1,124	1,754	1,201	964	2,120	2,722	2,347	1,916
July	--	--	--	--	--	--	--	1,348	1,019	1,536	1,180	1,342	951	701	1,173	1,740	1,389	2,157	2,312	2,972	2,987	3,166
Aug	--	--	--	--	--	--	--	884	999	1,521	1,528	274	1,059	370	475	776	1,704	943	1,733	1,792	2,747	2,174
Sept	--	--	--	--	--	--	--	167	208	--	--	--	--	--	--	--	--	--	--	--	--	--
TOTAL								3,942	2,986	4,150	4,949	2,308	3,052	1,803	2,772	4,270	4,514	4,699	7,343	9,147	8,884	8,733

Much of the early season angling targets Chinook Salmon, and occurs during a period in which the Area 3 and 4 stock composition is unknown or poorly resolved. As such, the Department of Fisheries and Oceans Canada (DFO) expressed interest in a May creel survey because genetic samples collected from Chinook Salmon would help them resolve stock composition and help determine exploitation rates on early-run-timing Chinook Salmon stocks. In 2019, during the month of May, 269 Chinook Salmon were biosampled, including 230 scale samples, and 22 tagged heads. Since then, 144-194 Chinook Salmon scale samples have been collected each May, representing 33-52% of the annual total. Since 2020, May accounted for 11-16 of the heads submitted to the Salmon Head Recovery Program, 26-56 % of the season-wide totals.

April Effort

Trailer counts made during April of 2019, 2020, 2023, and 2024 allowed the estimation of the amount of non-lodge angling activity that is being missed by starting the creel survey in May each year. Over these four years, April effort may have amounted to 766 to 1,303 trips, which represents an additional 8-11% of angling effort over that documented from May to August. It is not known what species these early season anglers are targeting.

Precision of the Results

Overall, our level of survey effort, both for CPUE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2024 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon, and Pacific Halibut), for which harvest estimates were precise to within 4-5%. Pink Salmon, Lingcod, and rockfish estimates were precise to within 11%, 7%, and 8%, respectively. Chum and Sockeye salmon estimates, and those for other groundfish, were not precise to within 15%, though the survey was not designed to meet any level of precision for these last taxa.

The precision of our harvest estimates depended largely on the variability in the CPUE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

Accuracy of the Results

The accuracy of the results depends on the quality of data provided by the anglers to the interviewers. This is partly why we have a target to sample 20% of estimated effort. Even so, each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 1,659 of the 2,494 Chinook Salmon (67%) and 5,203 of the 8,387 Coho Salmon (62%) reported as kept during angler interviews were recorded as observed by the interviewer, and 483 and 725 (29% and 14%) were handled for biosampling.

While not every fish can be observed², each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals. Angler compliance with requests to examine their catch is voluntary.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g., vessels outside of the survey area), as evidenced by surveys for which more lodge boats were operating than there were total boats observed in Area 3 during the flight. There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 27 and 28 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.
- We recommend that future surveys continue to run for four months (May to August) rather than three (June to August). May surveys will help document and quantify angling effort and harvest during that month. The Department of Fisheries and Oceans Canada (DFO) has expressed interest in a May creel survey because genetic samples collected from Chinook Salmon will help resolve the currently unknown early-season Area 3 and 4 stock composition, which will help determine exploitation rates on early-run-timing Chinook Salmon stocks.
- Trailer counts should be conducted in April 2025 for comparison with April trailer counts in 2019-2024; and should be continued throughout the creel survey season as in recent years.

² Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, boat launch traffic congestion causing tension, or COVID-19 concerns, among other reasons.

- Factors that are resulting increased Lingcod harvest should be investigated to inform managers should additional measures be enacted.

ACKNOWLEDGEMENTS

We would like to thank the creel surveyors (Anke De Wit, Grace Stewart, Audrey Parnell, Jay Byggdin, Kassidy Main-Cooper, and Rory Scott) for collecting large amounts of data and bio-samples from anglers. Thanks to Pacific Salmon Commission, who provided support for this work through the Northern Fund. Thanks to Chelsea May, Ryan Whitmore, Jessica Otley, Shaun Davies, and Darren Chow for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments and edits on previous drafts of this report. Last but not least, we thank the many anglers, charter operators, and fishing lodges who participated in the survey.

LITERATURE CITED

- Benjamini, Y., and Y. Hochberg. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B* 57: 289-300.
- Clark, W. G. 1992. Validation of the IPHC length-weight relationship for halibut. *International Pacific Halibut Commission Report of Assessment and Research Activities* 1991: 113-116.
- English, K. K., G. F. Searing, and D. A. Nagtegaal. 2002. Review of Georgia Strait recreational creel survey, 1983-1999. *Canadian Technical Report of Fisheries and Aquatic Sciences* 2414: 81 p.
- Goodman, L. 1960. On the exact variance of products. *Journal of the American Statistical Association* 55: 708-713.
- Robichaud, D., and A. Addison. 2018. North Coast (Areas 3 & 4) recreational angling creel survey 2017. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2019. North Coast (Areas 3 & 4) recreational angling creel survey 2018. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2020. North Coast (Areas 3 & 4) recreational angling creel survey 2019. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2021. North Coast (Areas 3 & 4) recreational angling creel survey 2020. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.

- Robichaud, D., and A. Addison. 2022. North Coast (Areas 3 & 4) recreational angling creel survey 2021. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2023. North Coast (Areas 3 & 4) recreational angling creel survey 2022. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., and A. Addison. 2024. North Coast (Areas 3 & 4) recreational angling creel survey 2023. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., A. Addison, and M. Imrie. 2016. North Coast (Areas 3 & 4) recreational angling creel survey 2015. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Robichaud, D., A. Addison, and M. Imrie. 2017. North Coast (Areas 3 & 4) recreational angling creel survey 2016. Report for North Coast - Skeena First Nations Stewardship Society, Prince Rupert, BC.
- Sokal, R.R. and F.J. Rohlf. 1995. Biometry. W. H. Freeman and Company, New York.
- Van Tongeren, V., and I. Winther. 2010. 2009 North Coast (Areas 3 & 4) creel survey statistics for salmon and groundfish. *Canadian Manuscript Report of Fisheries and Aquatic Sciences* 2907.
- Zar, J. H. 1984. Biostatistical Analysis. Prentice-Hall Inc., Englewood Cliffs, NJ.

APPENDICES

APPENDIX 1: DATA FORMS

2024 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE: _____ FLIGHT#: _____

SURVEYOR: _____

Note: only survey shaded subareas when conditions permit

Sub Area	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA STATE (1-4)
	SPORT (Fishing)	SPORT (Moving)	COMM. (Fishing)	COMM. (Moving)	S (F)	S (M)	C (F)	C (M)	START	FINISH	
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS:

SIGNATURE: _____

SHIFT SUMMARY SHEET

Landing Site: _____ Interviewer Name: _____

Interviewer Code: _____ Date: _____

Shift Code: _____ Shift Times: _____

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing

Totals:

--	--	--

Comments: _____

Trailer Count: _____ Time: _____

Database Tally # :

(office use only)

Page 52

APPENDIX 2: HPUE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER NON-LODGE BOAT TRIP) BY PERIOD, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon harvested per boat trip in May 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	1.21	0.29	2.50		0.64	2.81	4.00	3.43	3.54		1.33	2.75
Unguided	0.76	0.25	1.40	0.10	0.83	1.16	1.31	1.64	1.98	0.29	0.45	0.32
Combined	0.94	0.27	1.81	0.05	0.88	1.62	2.13		2.77	0.17	0.76	2.06
Chum Salmon												
Guided												
Unguided												
Combined												
Coho Salmon												
Guided	0.07		0.17		0.01	0.04		0.04	0.06			
Unguided												
Combined	0.03		0.06			0.01			0.03			
Pink Salmon												
Guided												
Unguided						0.01						
Combined						0.01						
Sockeye Salmon												
Guided												
Unguided												
Combined												
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in June 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.91	0.90	1.78		0.50	0.58	1.27	2.06	2.22	0.42	0.92	0.48
Unguided	0.45	1.26	0.59	0.01	0.45	0.70	1.15		1.33	0.12	0.35	0.25
Combined	0.68	1.06	1.00	0.00	0.46	0.67	1.18		1.77	0.27	0.63	0.48
Chum Salmon												
Guided					0.17							
Unguided		0.03										
Combined		0.01			0.01							
Coho Salmon												
Guided	0.36	0.98	2.11			2.09	0.20	1.70	1.95	0.96	1.48	0.78
Unguided	0.91	1.17	0.88		0.04	0.76	0.27		0.86	0.32	0.12	0.20
Combined	0.64	1.06	1.31		0.04	1.08	0.26		1.40	0.63	0.78	0.67
Pink Salmon												
Guided	0.27							0.11	0.12			
Unguided		0.14			0.02		0.03		0.04	0.04		0.02
Combined	0.14	0.06			0.02		0.03		0.08	0.02		
Sockeye Salmon												
Guided												
Unguided			0.06			0.01						
Combined			0.04			0.01						
Unknown Salmon												
Guided												
Unguided												
Combined												

Average number of salmon harvested per boat trip in July 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.44	0.58	1.00		0.33	0.33	0.96		0.50		0.28	0.39
Unguided	0.64	0.33	0.43			0.09	0.36	0.24	0.18		0.13	
Combined	0.54	0.46	0.58		0.02	0.16	0.46		0.31		0.18	0.27
Chum Salmon												
Guided			0.13									
Unguided						0.01						
Combined			0.03			0.01						
Coho Salmon												
Guided	6.04	4.72	8.25		2.00	3.78	8.44		8.37	1.92	2.38	1.78
Unguided	5.00	3.46	3.65		0.63	1.30	3.05	4.17	4.75	0.66	1.27	0.25
Combined	5.54	4.13	4.84		0.72	2.00	4.01		6.23	1.18	1.65	1.31
Pink Salmon												
Guided	0.67	0.13	0.13			0.48	0.16		0.32		0.16	
Unguided	0.44	0.42	0.22		0.10	0.32	0.33	0.38	0.41	0.23	0.33	
Combined	0.56	0.27	0.19		0.10	0.37	0.30		0.38	0.13	0.27	
Sockeye Salmon												
Guided		0.02							0.02			
Unguided						0.01		0.01	0.01			
Combined		0.01				0.01			0.01			
Unknown Salmon												
Guided												
Unguided								0.05	0.08			
Combined									0.05			

Average number of salmon harvested per boat trip in August 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	0.33	0.98	0.79		0.60	1.28	0.54	1.54	1.66	0.45	0.26	
Unguided	0.24	0.47	0.71		0.09	0.31	0.48	0.11	0.81	0.39	0.23	
Combined	0.28	0.75	0.63		0.13	0.57	0.49	0.09	1.21	0.41	0.24	
Chum Salmon												
Guided								0.01	0.01			
Unguided					0.01				0.01			
Combined					0.01				0.01			
Coho Salmon												
Guided	0.17	2.85	2.18		1.60	4.24	2.00	6.78	7.32	1.18	0.11	
Unguided	1.88	1.76	1.86		0.83	1.22	2.92	0.33	4.86	1.06	1.13	1.60
Combined	1.17	2.36	1.69		0.88	2.03	2.80	0.27	6.02	1.10	0.74	0.57
Pink Salmon												
Guided	0.08	0.20	0.16		0.20	0.45	0.08	0.39	0.43	0.09	0.05	
Unguided	0.06	0.18	0.21		0.13	0.16	0.34		0.45	0.11	0.10	
Combined	0.07	0.19	0.19		0.14	0.24	0.31		0.44	0.10	0.08	
Sockeye Salmon												
Guided						0.02		0.01	0.01			
Unguided						0.01	0.01					
Combined						0.01	0.01		0.00			
Unknown Salmon												
Guided												
Unguided							0.01				0.06	
Combined							0.01				0.04	

Average number of groundfish harvested per boat trip in May 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow-tooth Flounder												
Guided												
Unguided												
Combined												
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided	0.05	0.13						0.01	0.02			
Combined	0.03	0.07							0.01			
English Sole												
Guided												
Unguided								0.02	0.04			
Combined									0.02			
Greenlings												
Guided					0.02	0.07		0.03	0.04			0.58
Unguided					0.03		0.16	0.08	0.04		0.09	0.05
Combined					0.03	0.02	0.11		0.04		0.06	0.44
Lingcod												
Guided	2.50	5.00			0.09	0.44	0.07	0.74	0.63	0.80	2.50	2.92
Unguided	1.33	3.00	0.20		0.02	0.11	1.00	0.87	0.84	0.29	0.27	0.27
Combined	1.80	3.93	0.13		0.01	0.21	0.72	2.38	0.74	0.50	1.06	2.25
Longnose Skate												
Guided								0.01	0.02			
Unguided												
Combined									0.01			
Pacific Cod												
Guided	1.29		0.67					0.04	0.06	1.00	1.33	0.17
Unguided	0.05	0.13			0.03	0.11	0.06	0.41	0.67	0.29		0.09
Combined	0.54	0.07	0.25		0.03	0.08	0.04		0.36	0.58	0.47	0.13
Pacific Halibut												
Guided	2.86	3.86	1.50		0.21	1.00	1.00	1.39	1.29	3.20	3.17	2.17
Unguided	0.76	1.25	0.40		0.25	0.30	1.25	0.85	0.67	1.57	1.27	1.59
Combined	1.60	2.47	0.81		0.24	0.49	1.17	2.00	0.98	2.25	1.94	2.25
Pacific Rock Sole												
Guided								0.06	0.08			0.67
Unguided												
Combined									0.04			0.50
Pacific Spiny Dogfish												
Guided												
Unguided												
Combined												
Sablefish												
Guided												
Unguided												
Combined												
Spotted Ratfish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided					0.02							
Combined					0.01							
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of groundfish harvested per boat trip in June 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow-tooth Flounder												
Guided												
Unguided												
Combined												
Big Skate												
Guided												
Unguided						0.01						
Combined						0.01						
Cabezon												
Guided											0.04	
Unguided												
Combined											0.02	
English Sole												
Guided												
Unguided												
Combined												
Greenlings												
Guided			0.11			0.03						
Unguided						0.01				0.08		0.04
Combined			0.04			0.01				0.04		
Lingcod												
Guided		1.76				0.03		0.65	0.73	0.54	1.52	1.96
Unguided		1.11	0.18		0.02	0.10	0.37	0.73	0.59	0.96	0.31	0.58
Combined		1.47	0.12		0.02	0.09	0.30	0.71	0.66	0.76	0.90	1.67
Longnose Skate												
Guided												
Unguided												
Combined												
Pacific Cod												
Guided	0.45					0.39	0.07	0.07	0.07	0.79	0.52	0.30
Unguided		0.03			0.12	0.02		0.18	0.08	0.16	0.46	0.29
Combined	0.23	0.01			0.11	0.11	0.01	0.14	0.07	0.47	0.49	0.26
Pacific Halibut												
Guided	0.91	2.71	0.33			0.91	1.13	1.44	1.45	2.50	1.92	3.26
Unguided	0.09	0.86	1.00	0.01	0.31	0.64	0.80	0.91	0.88	0.64	0.96	0.89
Combined	0.50	1.87	0.77	0.00	0.30	0.71	0.86	1.29	1.16	1.55	1.43	3.07
Pacific Rock Sole												
Guided												
Unguided			0.06				0.05					
Combined			0.04				0.04					
Pacific Spiny Dogfish												
Guided												
Unguided												
Combined												
Sablefish												
Guided												
Unguided												
Combined												
Spotted Ratfish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided						0.01	0.02					
Combined						0.01	0.01					
Unknown Groundfish												
Guided												
Unguided							0.08					
Combined							0.07					

Average number of groundfish harvested per boat trip in July 2024, by trip type and subarea. Zeros hidden.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow-tooth Flounder												
Guided									0.01			
Unguided					0.02	0.01		0.01	0.01			
Combined					0.02	0.00			0.01			
Big Skate												
Guided						0.02						
Unguided												
Combined						0.00						
Cabezon												
Guided									0.01		0.03	
Unguided		0.10						0.00	0.00			
Combined		0.04							0.00		0.01	
English Sole												
Guided		0.02							0.01			
Unguided												
Combined		0.01							0.00			
Greenlings												
Guided		0.02							0.02			
Unguided							0.01	0.01	0.00	0.09		
Combined		0.01					0.01		0.01	0.05		
Lingcod												
Guided	0.41	3.40				0.29	0.32	3.00	0.96	1.16	0.88	1.28
Unguided	0.04	1.52	0.13		0.05	0.14	0.10	0.47	0.65	1.37	0.22	0.38
Combined	0.23	2.53	0.10		0.04	0.18	0.14	2.18	0.78	1.28	0.45	1.00
Longnose Skate												
Guided												
Unguided												
Combined												
Pacific Cod												
Guided	1.04	0.15				0.05			0.07		0.50	
Unguided	0.16	0.04			0.06	0.07	0.03	0.04	0.04	0.06	0.35	
Combined	0.62	0.10			0.05	0.06	0.02		0.05	0.03	0.40	
Pacific Halibut												
Guided	1.07	1.97	0.63		0.17	0.84	0.80	1.75	1.44	1.12	2.75	3.22
Unguided	0.88	1.85	0.78	0.01	0.57	0.71	0.84	0.96	1.01	1.26	1.05	2.88
Combined	0.98	1.91	0.74	0.01	0.54	0.75	0.84	1.55	1.19	1.20	1.64	3.12
Pacific Rock Sole												
Guided												
Unguided		0.02										
Combined		0.01										
Pacific Spiny Dogfish												
Guided												
Unguided	0.04					0.01						
Combined	0.02					0.00						
Sablefish												
Guided												
Unguided						0.01						
Combined						0.00						
Spotted Ratfish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided					0.03							
Combined					0.03							
Unknown Groundfish												
Guided												
Unguided						0.01		0.01	0.01			
Combined						0.01			0.00			

Average number of groundfish harvested per boat trip in August 2024, by trip type and subarea. Zeros hidden.

Species / Trip Type	Subarea											
	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow-tooth Flounder												
Guided								0.01	0.01			
Unguided												
Combined									0.00			
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided												
Combined												
English Sole												
Guided												
Unguided					0.03							
Combined					0.03							
Greenlings												
Guided		0.05	0.04					0.01	0.01			
Unguided		0.06			0.04				0.01			
Combined		0.05			0.04				0.01			
Lingcod												
Guided	0.08	2.32	1.75			0.45	1.38	0.77	0.72		0.68	1.33
Unguided		0.94			0.01	0.03	0.08	1.11	0.24	0.39	0.45	2.40
Combined	0.03	1.69			0.01	0.14	0.26	0.91	0.46	0.24	0.54	1.71
Longnose Skate												
Guided												
Unguided												
Combined												
Pacific Cod												
Guided	2.50	0.39	0.82			0.05	0.08	0.02	0.01	1.18	1.84	0.56
Unguided	0.24	0.09	0.43		0.01	0.22	0.16	0.56	0.09		0.03	
Combined	1.17	0.25	0.38		0.01	0.18	0.15	0.45	0.05	0.45	0.72	0.36
Pacific Halibut												
Guided	2.17	3.20	2.98		0.20	1.24	1.31	1.29	1.26	0.91	3.32	3.33
Unguided	1.29	1.47	0.43		0.48	0.92	0.86	2.56	1.05	0.78	1.42	1.20
Combined	1.66	2.41	0.38		0.46	1.01	0.92	2.73	1.15	0.83	2.14	2.57
Pacific Rock Sole												
Guided												
Unguided												
Combined												
Pacific Spiny Dogfish												
Guided												
Unguided					0.01	0.01						
Combined					0.01	0.00						
Sablefish												
Guided												
Unguided												
Combined												
Spotted Ratfish												
Guided												
Unguided												
Combined												
Starry Flounder												
Guided												
Unguided												
Combined												
Unknown Groundfish												
Guided												
Unguided												
Combined												

Average number of rockfish harvested per boat trip in May 2024 by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.43	0.86			0.01	0.04	0.21	0.31	0.33		2.67	1.00
Unguided	0.24	0.63			0.22	0.01	0.06	0.11	0.16			
Combined	0.31	0.73			0.21	0.02	0.11	0.25	0.24		0.94	0.75
Boccacio Rockfish												
Guided					0.02	0.07						
Unguided												
Combined					0.01	0.02						
Canary Rockfish												
Guided					0.01			0.01	0.02			
Unguided	0.10	0.25			0.02							
Combined	0.06	0.13			0.03				0.01			
China Rockfish												
Guided	0.07		0.17		0.01	0.04		0.03	0.02			
Unguided	0.05	0.13					0.09	0.06	0.04	0.14	0.09	0.09
Combined	0.06	0.07	0.06			0.01	0.07	0.13	0.03	0.08	0.06	
Copper Rockfish												
Guided	0.29	0.43	0.17		0.01	0.04	0.43	0.13	0.06			
Unguided					0.03		0.13	0.05			0.27	0.14
Combined	0.11	0.20	0.06		0.03	0.01	0.22		0.03		0.18	
Dusky Rockfish												
Guided								0.01	0.02			
Unguided												
Combined									0.01			
Quillback Rockfish												
Guided	0.50	1.00			0.02	0.04	0.07	0.16	0.15			0.08
Unguided	0.52	1.38			0.12	0.06	0.13	0.15	0.18	0.14		0.05
Combined	0.51	1.20			0.13	0.05	0.11	0.25	0.17	0.08		0.06
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided								0.07	0.10			
Unguided												
Combined									0.05			
Widow Rockfish												
Guided												
Unguided							0.03	0.01				
Combined							0.02					
Yelloweye Rockfish												
Guided												
Unguided	0.19	0.50										
Combined	0.11	0.27										
Yellowtail Rockfish												
Guided	0.64		1.50					0.04	0.06			
Unguided												
Combined	0.26		0.56						0.03			
Unknown rockfish												
Guided							0.14	0.16	0.17			1.25
Unguided	0.19		0.40	0.05	0.37	0.01	1.00	0.40	0.06		0.18	0.09
Combined	0.11		0.25	0.03	0.35	0.01	0.74		0.12		0.12	0.94

Average number of rockfish harvested per boat trip in June 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided		0.40						0.11	0.12	0.50	0.52	0.39
Unguided	0.18	0.37			0.02	0.16	0.20	3.09	0.13		0.04	0.02
Combined	0.09	0.39			0.02	0.12	0.16	2.50	0.13	0.24	0.27	0.33
Boccacio Rockfish												
Guided												
Unguided							0.02		0.01			
Combined							0.01		0.00			
Canary Rockfish												
Guided		0.02	0.33			0.06						
Unguided												
Combined		0.01	0.12			0.01						
China Rockfish												
Guided		0.07				0.03				0.04		
Unguided	0.27								0.03			
Combined	0.14	0.04				0.01			0.01	0.02		
Copper Rockfish												
Guided		0.07	0.22					0.03	0.04			
Unguided						0.05	0.08		0.02			
Combined		0.04	0.08			0.04	0.07		0.03			
Dusky Rockfish												
Guided												
Unguided						0.01			0.07			
Combined						0.01			0.04			
Quillback Rockfish												
Guided	0.09	0.31			0.17	0.03	0.47	0.24	0.20	0.46		
Unguided		0.14			0.06	0.13	0.08	0.09	0.17	0.12	0.04	0.09
Combined	0.05	0.23			0.07	0.11	0.16	0.14	0.18	0.29	0.02	0.04
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided									0.01	0.08		0.04
Combined									0.00	0.04		
Tiger Rockfish												
Guided						0.06		0.02	0.02			
Unguided						0.02			0.01			
Combined						0.03			0.01			
Widow Rockfish												
Guided												
Unguided					0.03							
Combined					0.03							
Yelloweye Rockfish												
Guided												
Unguided												
Combined												
Yellowtail Rockfish												
Guided		0.05	0.89			0.03	0.07	0.02	0.02			
Unguided					0.22	0.02			0.03		0.31	0.15
Combined		0.03	0.31		0.21	0.02	0.01		0.02		0.16	
Unknown rockfish												
Guided	0.09	0.05						0.02	0.03		0.28	0.39
Unguided	0.18					0.03	0.69		0.01	0.12	0.04	0.07
Combined	0.14	0.03				0.02	0.55		0.02	0.06	0.16	0.33

Average number of rockfish harvested per boat trip in July 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided	0.07	0.53					0.32	2.63	0.51			
Unguided	0.24	0.42			0.14	0.07	0.12	0.22	0.26	0.34	0.13	2.38
Combined	0.15	0.48			0.13	0.05	0.16	1.91	0.37	0.20	0.09	0.73
Boccacio Rockfish												
Guided												
Unguided							0.02	0.01	0.01		0.02	
Combined							0.01		0.00		0.01	
Canary Rockfish												
Guided						0.02			0.01			
Unguided					0.01	0.01	0.01	0.00				
Combined					0.01	0.01	0.01		0.00			
China Rockfish												
Guided		0.08							0.05		0.06	
Unguided		0.10			0.01	0.02	0.01	0.03	0.04		0.02	
Combined		0.09			0.01	0.01	0.01		0.04		0.03	
Copper Rockfish												
Guided		0.02				0.02	0.20		0.03		0.09	
Unguided		0.04		0.01	0.06		0.07	0.03	0.02		0.03	
Combined		0.03		0.00	0.05	0.00	0.09		0.02		0.05	
Dusky Rockfish												
Guided		0.03										
Unguided											0.03	
Combined		0.02									0.02	
Quillback Rockfish												
Guided	0.11	0.42				0.17	0.28	0.50	0.33	0.28	0.09	
Unguided	0.20	0.17			0.15	0.07	0.08	0.14	0.17	0.06	0.15	0.25
Combined	0.15	0.30			0.14	0.10	0.11	0.36	0.24	0.15	0.13	0.08
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided	0.04	0.03					0.04	0.38	0.04	0.08		
Unguided		0.08					0.02	0.03	0.04	0.11	0.02	
Combined	0.02	0.05					0.02	0.27	0.04	0.10	0.01	
Widow Rockfish												
Guided									0.01			
Unguided						0.01		0.01	0.01			
Combined						0.00			0.01			
Yelloweye Rockfish												
Guided												
Unguided					0.01							
Combined					0.01							
Yellowtail Rockfish												
Guided		0.07					0.16		0.03		0.03	
Unguided					0.05	0.07	0.01	0.03	0.03		0.02	
Combined		0.04			0.04	0.05	0.04		0.03		0.02	
Unknown rockfish												
Guided	0.07	0.40							0.09		0.09	
Unguided	0.12	0.15			0.10		0.08	0.09	0.10	0.49	0.13	
Combined	0.10	0.29			0.10		0.06		0.10	0.28	0.12	

Average number of rockfish harvested per boat trip in August 2024, by trip type and subarea. Zeros hidden.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Black Rockfish												
Guided		1.15	0.84			0.09	0.46	0.19	0.16		0.11	
Unguided	0.06	0.29			0.23	0.06	0.11	0.22	0.17		0.16	
Combined	0.03	0.76			0.21	0.07	0.15	0.18	0.17		0.14	
Boccacio Rockfish												
Guided		0.29	0.21									
Unguided												
Combined		0.16										
Canary Rockfish												
Guided												
Unguided							0.01					
Combined							0.01					
China Rockfish												
Guided		0.12	0.09				0.08	0.01				
Unguided	0.06	0.03				0.01			0.01			
Combined	0.03	0.08				0.01	0.01		0.00			
Copper Rockfish												
Guided		0.02	0.02			0.19		0.02	0.02			
Unguided						0.01	0.07				0.03	
Combined		0.01				0.06	0.06		0.01		0.02	
Dusky Rockfish												
Guided												
Unguided			0.14									
Combined			0.13									
Quillback Rockfish												
Guided	0.33	0.59	0.50			0.17	0.23	0.21	0.21		0.37	0.11
Unguided		0.12				0.08	0.06	0.56	0.11		0.13	
Combined	0.14	0.37				0.10	0.08	0.45	0.15		0.22	0.07
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided		0.07	0.05				0.08	0.05	0.04		0.05	
Unguided						0.04		0.11	0.01			
Combined		0.04				0.03	0.01	0.09	0.02		0.02	
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided		0.05	0.04									
Unguided												
Combined		0.03										
Yellowtail Rockfish												
Guided		0.02	0.02			0.02	0.92	0.09	0.01			
Unguided					0.05	0.03			0.01			
Combined		0.01			0.05	0.02	0.12		0.01			
Unknown rockfish												
Guided		0.02	0.02			0.31	0.15	0.06	0.05		0.37	0.78
Unguided		0.26	0.14		0.11	0.03	0.05		0.05	0.17	0.29	
Combined		0.13	0.13		0.10	0.10	0.06		0.05	0.10	0.32	0.50

APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2024, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded. Also shown, the number of lodge boats that would have been visible in Area 3 during flight.

Month	Date	Subarea											Area 3	
		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Lodge Boats
May	2 May 2024	2	0	1	1	2	5	0	2	3	1	0	0	0
	5 May 2024	1	1	0	2	2	2	2	0	3	0	0	0	0
	11 May 2024	4	5	3	2	13	11	0	0	17	2	2	0	0
	15 May 2024	0	2	2	1	4	1	1	0	3	2	0	0	4
	16 May 2024	1	1	4	2	1	4	0	0	0	0	2	0	4
	22 May 2024	1	3	3	1	5	8	3	1	7	4	1	5	4
	25 May 2024	3	1	2	1	2	10	3	0	9	0	4	0	4
	28 May 2024	2	9	4	1	5	5	3	3	7	1	1	0	7
June	2 Jun 2024	2	3	14	0	3	2	4	0	7	0	0	na	10
	9 Jun 2024	9	7	2	1	11	4	8	0	20	5	1	0	11
	12 Jun 2024	3	10	7	2	3	7	3	2	15	7	0	2	8
	18 Jun 2024	7	4	2	1	4	8	3	0	5	4	4	0	11
	20 Jun 2024	2	3	13	0	3	5	5	0	11	8	1	2	10
	23 Jun 2024	3	14	5	1	2	5	5	2	8	8	1	na	11
	25 Jun 2024	7	7	5	7	1	7	3	3	8	17	2	3	5
	26 Jun 2024	2	10	3	0	1	8	0	3	10	8	1	1	5
July	7 Jul 2024	9	3	11	3	4	25	3	0	7	6	8	1	7
	9 Jul 2024	12	9	3	2	2	9	3	0	7	2	6	0	11
	10 Jul 2024	3	5	1	0	4	6	1	1	4	0	0	na	8
	13 Jul 2024	8	6	5	4	4	17	1	0	32	0	10	0	8
	18 Jul 2024	6	12	9	0	4	10	4	0	17	2	4	na	11
	20 Jul 2024	10	33	9	5	6	19	7	5	26	19	4	0	11
	23 Jul 2024	2	14	15	0	3	2	15	0	24	5	0	1	10
August	1 Aug 2024	5	22	2	5	3	22	11	4	21	9	6	1	11
	11 Aug 2024	10	9	8	1	3	17	6	0	11	4	8	0	10
	15 Aug 2024	6	13	8	1	2	13	0	0	13	4	6	0	7
	22 Aug 2024	10	8	3	1	1	3	2	0	6	5	5	na	4
	27 Aug 2024	5	6	5	0	2	13	1	0	8	3	5	0	3
	29 Aug 2024	1	14	6	1	7	4	2	0	8	5	3	na	3
	31 Aug 2024	5	19	10	0	4	11	2	3	14	7	5	1	3

APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA, AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	67.3	25.5	145.1	4.5	230.8	575.0	204.3	0	998.5	14.6	56.4	110.2	2,432
Lodge Report	45.8	17.4	98.8	0	0	0	0	0	0	0	0	0	162
MAY RETAINED	113.2	42.9	243.9	4.5	230.8	575.0	204.3	0	998.5	14.6	56.4	110.2	2,594
SE	14.5	8.2	34.2	1.7	36.7	51.6	27.0	0	64.3	12.8	25.5	44.2	114
June													
Creel Estimate	96.8	242.9	201.6	0.4	92.1	247.9	263.7	0	1114.7	118.4	51.9	36.9	2,467
Lodge Report	68.1	171.0	141.9	0	0	0	0	0	0	0	0	0	381
JUNE RETAINED	164.9	413.9	343.5	0.4	92.1	247.9	263.7	0	1114.7	118.4	51.9	36.9	2,848
SE	24.8	36.1	54.2	0.3	13.2	39.1	30.2	0	63.8	37.3	13.3	12.5	115
July													
Creel Estimate	123.7	192.1	161.2	0	4.7	98.0	141.1	0	285.3	0	42.4	7.2	1,056
Lodge Report	56.0	87.0	73.0	0	0	0	0	0	0	0	0	0	216
JULY RETAINED	179.7	279.0	234.1	0	4.7	98.0	141.1	0	285.3	0	42.4	7.2	1,272
SE	27.4	27.5	43.9	0	3.4	12.1	17.8	0	23.9	0	13.8	3.7	68
August													
Creel Estimate	76.6	464.4	185.6	0	23.4	419.1	100.1	5.8	849.8	133.3	80.7	0	2,339
Lodge Report	19.6	118.9	47.5	0	0	0	0	0	0	0	0	0	186
AUGUST RETAINED	96.3	583.3	233.1	0	23.4	419.1	100.1	5.8	849.8	133.3	80.7	0	2,525
SE	22.3	75.0	57.2	0	7.6	46.2	14.2	4.1	67.0	42.0	21.6	0	136
TOTAL													
Creel Estimate	364	925	693	5	351	1,340	709	6	3,248	266	231	154	8,294
Lodge Report	190	394	361	0	0	0	0	0	0	0	0	0	945
TOTAL RETAINED	554	1,319	1,055	5	351	1,340	709	6	3,248	266	231	154	9,239
SE	46	88	96	2	40	80	46	4	115	58	39	46	222

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	38.8	12.8	85.1	3.6	200.0	344.3	68.8	0	578.1	14.6	0	0	1,346
Lodge Report	3.1	1.0	6.9	0	0	0	0	0	0	0	0	0	11
MAY RELEASED	41.9	13.8	91.9	3.6	200.0	344.3	68.8	0	578.1	14.6	0	0	1,357
SE	10.6	6.3	26.3	1.4	33.6	48.7	13.1	0	66.0	12.8	0	0	95
June													
Creel Estimate	109.7	376.2	930.4	0.4	168.8	434.5	218.2	0	929.9	473.7	58.4	19.8	3,720
Lodge Report	45.1	154.9	383.0	0	0	0	0	0	0	0	0	0	583
JUNE RELEASED	154.8	531.1	1313.4	0.4	168.8	434.5	218.2	0	929.9	473.7	58.4	19.8	4,303
SE	47.2	63.4	219.6	0.3	29.5	68.2	34.8	0	67.4	173.5	15.4	10.4	310
July													
Creel Estimate	327.0	325.0	447.7	0.5	80.1	428.6	167.2	10.7	572.8	111.2	144.7	3.1	2,618
Lodge Report	242.3	240.9	331.8	0	0	0	0	0	0	0	0	0	815
JULY RELEASED	569.3	565.9	779.4	0.5	80.1	428.6	167.2	10.7	572.8	111.2	144.7	3.1	3,433
SE	55.3	60.7	112.8	0.4	26.7	54.7	26.0	8.4	57.0	34.4	30.5	2.5	171
August													
Creel Estimate	28.7	190.7	259.8	0.3	9.4	402.1	70.9	0	201.2	77.7	60.5	0	1,301
Lodge Report	12.6	83.6	113.8	0	0	0	0	0	0	0	0	0	210
AUGUST RELEASED	41.3	274.3	373.6	0.3	9.4	402.1	70.9	0	201.2	77.7	60.5	0	1,511
SE	15.4	51.5	91.9	0.3	5.5	56.6	24.6	0	25.9	28.2	19.4	0	130
TOTAL													
Creel Estimate	504	905	1,723	5	458	1,609	525	11	2,282	677	264	23	8,986
Lodge Report	303	480	835	0	0	0	0	0	0	0	0	0	1,619
TOTAL RELEASED	807	1,385	2,558	5	458	1,609	525	11	2,282	677	264	23	10,605
SE	75	102	265	2	52	115	52	8	113	180	39	11	389

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	2.0	0	5.0	0	0	3.7	0	0	10.5	0	0	0	21
Lodge Report	1.7	0	4.3	0	0	0	0	0	0	0	0	0	6
MAY RETAINED	3.8	0	9.3	0	0	3.7	0	0	10.5	0	0	0	27
SE	1.4	0	3.9	0	0	2.7	0	0	5.5	0	0	0	7
June													
Creel Estimate	90.3	242.9	263.6	0	7.7	399.8	57.6	0	880.2	282.4	64.8	51.0	2,340
Lodge Report	70.8	190.5	206.7	0	0	0	0	0	0	0	0	0	468
JUNE RETAINED	161.1	433.4	470.3	0	7.7	399.8	57.6	0	880.2	282.4	64.8	51.0	2,808
SE	26.9	48.8	69.3	0	2.7	66.2	13.3	0	83.1	99.3	25.4	19.1	174
July													
Creel Estimate	1272.5	1710.1	1343.0	0	155.5	1261.3	1220.2	0	5720.9	282.0	379.1	34.9	13,379
Lodge Report	666.0	895.1	702.9	0	0	0	0	0	0	0	0	0	2,264
JULY RETAINED	1938.5	2605.1	2045.9	0	155.5	1261.3	1220.2	0	5720.9	282.0	379.1	34.9	15,643
SE	118.8	141.2	200.9	0	30.6	104.7	93.6	0	171.1	81.4	70.4	15.1	369
August													
Creel Estimate	325.7	1467.9	501.0	0	164.0	1492.4	571.5	17.4	4232.7	355.3	248.9	13.7	9,391
Lodge Report	196.5	885.3	302.2	0	0	0	0	0	0	0	0	0	1,384
AUGUST RETAINED	522.2	2353.3	803.2	0	164.0	1492.4	571.5	17.4	4232.7	355.3	248.9	13.7	10,775
SE	101.2	185.6	122.6	0	28.3	116.1	51.4	8.8	158.0	84.3	75.0	9.8	338
TOTAL													
Creel Estimate	1,691	3,421	2,113	0	327	3,157	1,849	17	10,844	920	693	100	25,131
Lodge Report	935	1,971	1,216	0	0	0	0	0	0	0	0	0	4,122
TOTAL RETAINED	2,626	5,392	3,329	0	327	3,157	1,849	17	10,844	920	693	100	29,253
SE	158	238	245	0	42	170	108	9	247	154	106	26	530

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	10.5	0	0	0	11
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	10.5	0	0	0	11
SE	0	0	0	0	0	0	0	0	7.4	0	0	0	7
June													
Creel Estimate	0	82.9	217.1	0	3.8	26.7	3.0	0	49.7	18.2	1.6	0	403
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	82.9	217.1	0	3.8	26.7	3.0	0	49.7	18.2	1.6	0	403
SE	0	31.6	116.9	0	2.7	14.2	2.1	0	15.8	10.6	1.4	0	123
July													
Creel Estimate	154.6	195.8	197.0	0	4.7	110.2	267.0	42.9	595.2	11.9	2.5	0	1,582
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	154.6	195.8	197.0	0	4.7	110.2	267.0	42.9	595.2	11.9	2.5	0	1,582
SE	78.9	40.8	74.2	0	3.4	27.8	50.5	33.5	101.6	8.6	1.8	0	168
August													
Creel Estimate	28.7	174.2	37.1	0	16.4	184.0	150.2	46.3	333.8	44.4	0	0	1,015
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	28.7	174.2	37.1	0	16.4	184.0	150.2	46.3	333.8	44.4	0	0	1,015
SE	21.0	70.6	26.3	0	6.1	40.6	33.2	22.8	43.7	25.8	0	0	110
TOTAL													
Creel Estimate	183	453	451	0	25	321	420	89	989	75	4	0	3,011
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	183	453	451	0	25	321	420	89	989	75	4	0	3,011
SE	82	87	141	0	8	51	60	41	112	29	2	0	236

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	3.7	0	0	0	0	0	0	4
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	3.7	0	0	0	0	0	0	4
SE	0	0	0	0	0	2.7	0	0	0	0	0	0	3
June													
Creel Estimate	19.4	14.8	0	0	3.8	0	6.1	0	52.4	9.1	0	0	106
Lodge Report	1.0	0.8	0	0	0	0	0	0	0	0	0	0	2
JUNE RETAINED	20.4	15.6	0	0	3.8	0	6.1	0	52.4	9.1	0	0	107
SE	10.7	7.4	0	0	1.9	0	3.0	0	11.8	7.6	0	0	19
July													
Creel Estimate	128.1	110.8	53.7	0	21.2	232.7	91.2	0	345.9	31.8	62.4	0	1,078
Lodge Report	6.9	6.0	2.9	0	0	0	0	0	0	0	0	0	16
JULY RETAINED	135.1	116.8	56.6	0	21.2	232.7	91.2	0	345.9	31.8	62.4	0	1,094
SE	36.2	24.2	17.4	0	6.9	31.8	20.6	0	37.1	13.2	16.7	0	74
August													
Creel Estimate	19.2	116.1	55.7	0	25.8	173.8	62.6	0	307.7	33.3	26.9	0	821
Lodge Report	1.0	6.3	3.0	0	0	0	0	0	0	0	0	0	10
AUGUST RETAINED	20.2	122.4	58.7	0	25.8	173.8	62.6	0	307.7	33.3	26.9	0	831
SE	9.7	45.2	28.6	0	8.5	27.0	13.7	0	40.9	18.2	11.8	0	78
TOTAL													
Creel Estimate	167	242	109	0	51	410	160	0	706	74	89	0	2,008
Lodge Report	9	13	6	0	0	0	0	0	0	0	0	0	28
TOTAL RETAINED	176	255	115	0	51	410	160	0	706	74	89	0	2,036
SE	39	52	33	0	11	42	25	0	56	24	20	0	109

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	41.5	0	0	0	2.7	6.1	0	16.6	0	0	0	67
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	41.5	0	0	0	2.7	6.1	0	16.6	0	0	0	67
SE	0	19.5	0	0	0	2.2	3.0	0	5.7	0	0	0	21
July													
Creel Estimate	481.6	1189.3	1262.4	0	108.3	921.5	744.7	0	3263.6	99.3	167.1	5.1	8,243
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	481.6	1189.3	1262.4	0	108.3	921.5	744.7	0	3263.6	99.3	167.1	5.1	8,243
SE	72.5	176.4	325.5	0	28.3	127.7	80.4	0	195.9	27.3	46.4	4.2	455
August													
Creel Estimate	661.1	779.6	222.7	0	72.6	923.4	450.5	17.4	2322.3	199.9	174.9	0	5,824
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	661.1	779.6	222.7	0	72.6	923.4	450.5	17.4	2322.3	199.9	174.9	0	5,824
SE	238.3	134.0	93.1	0	21.6	119.8	65.3	12.3	168.7	73.6	57.0	0	374
TOTAL													
Creel Estimate	1,143	2,010	1,485	0	181	1,848	1,201	17	5,602	299	342	5	14,134
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	1,143	2,010	1,485	0	181	1,848	1,201	17	5,602	299	342	5	14,134
SE	249	222	339	0	36	175	104	12	259	78	73	4	589

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	3.0	0	0	1.9	0	0	0	0	0	0	0	5
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	3.0	0	0	1.9	0	0	0	0	0	0	0	5
SE	0	2.3	0	0	1.4	0	0	0	0	0	0	0	3
July													
Creel Estimate	0	0	9.0	0	0	6.1	0	0	0	0	0	0	15
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	9.0	0	0	6.1	0	0	0	0	0	0	15
SE	0	0	6.5	0	0	4.4	0	0	0	0	0	0	8
August													
Creel Estimate	0	0	0	0	2.3	0	0	0	4.7	0	0	0	7
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	2.3	0	0	0	4.7	0	0	0	7
SE	0	0	0	0	1.7	0	0	0	2.5	0	0	0	3
TOTAL													
Creel Estimate	0	3	9	0	4	6	0	0	5	0	0	0	27
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	3	9	0	4	6	0	0	5	0	0	0	27
SE	0	2	7	0	2	4	0	0	2	0	0	0	9

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	25.8	8.9	0	0	0	5.3	0	0	55.2	36.4	8.1	0	140
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	25.8	8.9	0	0	0	5.3	0	0	55.2	36.4	8.1	0	140
SE	13.4	5.0	0	0	0	4.4	0	0	12.9	30.4	5.6	0	37
July													
Creel Estimate	4.4	0	17.9	0	0	12.2	0	0	27.0	0	0	0	62
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	4.4	0	17.9	0	0	12.2	0	0	27.0	0	0	0	62
SE	3.1	0	13.1	0	0	5.3	0	0	8.2	0	0	0	17
August													
Creel Estimate	0	24.9	0	0	0	13.6	2.1	0	7.1	11.1	0	0	59
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	24.9	0	0	0	13.6	2.1	0	7.1	11.1	0	0	59
SE	0	13.4	0	0	0	6.0	1.5	0	3.0	8.2	0	0	17
TOTAL													
Creel Estimate	30	34	18	0	0	31	2	0	89	48	8	0	260
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	30	34	18	0	0	31	2	0	89	48	8	0	260
SE	14	14	13	0	0	9	2	0	16	31	6	0	44

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	7.8	0	0	2.7	0	0	0	0	0	0	10
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	7.8	0	0	2.7	0	0	0	0	0	0	10
SE	0	0	5.8	0	0	2.2	0	0	0	0	0	0	6
July													
Creel Estimate	0	3.7	0	0	0	6.1	0	0	13.5	0	0	0	23
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	3.7	0	0	0	6.1	0	0	13.5	0	0	0	23
SE	0	2.6	0	0	0	4.4	0	0	5.1	0	0	0	7
August													
Creel Estimate	0	0	0	0	0	6.8	2.1	0	2.4	0	0	0	11
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	6.8	2.1	0	2.4	0	0	0	11
SE	0	0	0	0	0	3.5	1.5	0	1.7	0	0	0	4
TOTAL													
Creel Estimate	0	4	8	0	0	16	2	0	16	0	0	0	45
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	4	8	0	0	16	2	0	16	0	0	0	45
SE	0	3	6	0	0	6	2	0	5	0	0	0	10

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	6.5	0	6.7	0	0	0	13
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	6.5	0	6.7	0	0	0	13
SE	0	0	0	0	0	0	5.3	0	2.8	0	0	0	6
August													
Creel Estimate	9.6	8.3	0	0	0	0	2.1	0	0	0	0	0	20
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	9.6	8.3	0	0	0	0	2.1	0	0	0	0	0	20
SE	7.0	6.0	0	0	0	0	1.5	0	0	0	0	0	9
TOTAL													
Creel Estimate	10	8	0	0	0	0	9	0	7	0	0	0	33
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	10	8	0	0	0	0	9	0	7	0	0	0	33
SE	7	6	0	0	0	0	6	0	3	0	0	0	11

Estimated numbers of unknown salmon retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	0	0	0	42.7	0	0	0	43
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	0	0	0	0	0	0	42.7	0	0	0	43
SE	0	0	0	0	0	0	0	0	21.9	0	0	0	22
August													
Creel Estimate	0	0	0	0	0	0	2.1	0	0	0	13.5	0	16
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	2.1	0	0	0	13.5	0	16
SE	0	0	0	0	0	0	1.5	0	0	0	9.8	0	10
TOTAL													
Creel Estimate	0	0	0	0	0	0	2	0	43	0	13	0	58
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	0	0	0	0	0	2	0	43	0	13	0	58
SE	0	0	0	0	0	0	2	0	22	0	10	0	24

Estimated numbers of unknown salmon released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
July													
Creel Estimate	0	0	0	0	0	73.5	0	0	24.7	0	0	0	98
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	73.5	0	0	24.7	0	0	0	98
SE	0	0	0	0	0	52.3	0	0	11.7	0	0	0	54
August													
Creel Estimate	0	0	92.8	0	0	17.0	2.1	0	0	0	0	0	112
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	92.8	0	0	17.0	2.1	0	0	0	0	0	112
SE	0	0	65.7	0	0	12.4	1.5	0	0	0	0	0	67
TOTAL													
Creel Estimate	0	0	93	0	0	91	2	0	25	0	0	0	210
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	93	0	0	91	2	0	25	0	0	0	210
SE	0	0	66	0	0	54	2	0	12	0	0	0	86

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	114.3	236.1	65.0	0	61.5	175.8	112.6	125.8	353.9	197.4	143.3	120.2	1,706
Lodge Report	24.5	50.6	13.9	0	0	0	0	0	0	0	0	0	89
MAY RETAINED	138.7	286.7	79.0	0	61.5	175.8	112.6	125.8	353.9	197.4	143.3	120.2	1,795
SE	16.0	37.1	20.6	0	17.9	31.9	15.7	47.2	36.7	43.5	26.7	21.7	101
June													
Creel Estimate	71.0	426.6	155.1	0.4	59.5	261.2	194.0	100.9	731.2	692.4	118.3	235.4	3,046
Lodge Report	35.8	215.0	78.2	0	0	0	0	0	0	0	0	0	329
JUNE RETAINED	106.7	641.6	233.2	0.4	59.5	261.2	194.0	100.9	731.2	692.4	118.3	235.4	3,375
SE	28.0	40.5	36.7	0.3	10.6	35.5	25.8	28.4	51.2	113.4	17.3	30.8	153
July													
Creel Estimate	225.3	790.4	205.9	0.5	117.8	471.4	254.0	60.7	1089.4	286.0	376.6	83.1	3,961
Lodge Report	78.2	274.3	71.5	0	0	0	0	0	0	0	0	0	424
JULY RETAINED	303.5	1064.7	277.4	0.5	117.8	471.4	254.0	60.7	1089.4	286.0	376.6	83.1	4,385
SE	32.1	63.1	55.7	0.4	16.7	47.5	36.6	15.7	60.4	38.7	33.9	8.5	136
August													
Creel Estimate	459.9	1501.1	111.3	0	86.7	742.8	187.7	173.7	807.2	266.5	719.7	61.6	5,118
Lodge Report	63.0	205.7	15.3	0	0	0	0	0	0	0	0	0	284
AUGUST RETAINED	522.9	1706.8	126.6	0	86.7	742.8	187.7	173.7	807.2	266.5	719.7	61.6	5,402
SE	67.0	119.0	57.2	0	15.2	59.7	23.4	28.0	76.8	70.3	65.8	9.5	206
TOTAL													
Creel Estimate	870	2,954	537	1	325	1,651	748	461	2,982	1,442	1,358	500	13,831
Lodge Report	201	746	179	0	0	0	0	0	0	0	0	0	1,126
TOTAL RETAINED	1,072	3,700	716	1	325	1,651	748	461	2,982	1,442	1,358	500	14,957
SE	81	145	90	1	31	90	53	64	116	146	81	40	308

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	149.0	153.2	125.1	0	7.7	161.2	35.4	47.2	308.3	241.3	86.8	76.8	1,392
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	149.0	153.2	125.1	0	7.7	161.2	35.4	47.2	308.3	241.3	86.8	76.8	1,392
SE	39.5	39.3	50.5	0	5.6	36.8	11.8	35.5	60.9	106.3	26.2	26.6	158
June													
Creel Estimate	0	319.9	62.0	0	19.2	215.9	66.7	100.9	491.1	892.8	63.2	127.6	2,359
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	319.9	62.0	0	19.2	215.9	66.7	100.9	491.1	892.8	63.2	127.6	2,359
SE	0	57.2	29.9	0	9.7	75.4	18.3	35.3	58.3	244.2	21.3	28.0	275
July													
Creel Estimate	167.9	856.9	143.3	0	108.3	275.5	86.8	200.0	680.6	127.1	503.8	84.2	3,234
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	167.9	856.9	143.3	0	108.3	275.5	86.8	200.0	680.6	127.1	503.8	84.2	3,234
SE	48.6	164.9	61.6	0	34.3	84.6	29.4	94.4	75.9	36.0	88.7	22.6	259
August													
Creel Estimate	450.3	1235.7	0	0	23.4	310.1	89.7	57.9	466.3	122.2	578.5	44.5	3,379
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	450.3	1235.7	0	0	23.4	310.1	89.7	57.9	466.3	122.2	578.5	44.5	3,379
SE	166.4	184.1	0	0	7.1	57.9	22.4	26.1	70.9	48.1	160.5	12.7	315
TOTAL													
Creel Estimate	767	2,566	330	0	159	963	279	406	1,946	1,383	1,232	333	10,364
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	767	2,566	330	0	159	963	279	406	1,946	1,383	1,232	333	10,364
SE	178	257	85	0	37	133	43	110	134	273	186	46	517

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	128.6	376.6	10.0	0	3.8	73.3	68.8	149.4	266.3	43.9	78.1	120.2	1,319
Lodge Report	39.4	115.5	3.1	0	0	0	0	0	0	0	0	0	158
MAY RETAINED	168.0	492.1	13.1	0	3.8	73.3	68.8	149.4	266.3	43.9	78.1	120.2	1,477
SE	29.8	81.1	7.8	0	2.8	34.0	21.0	89.2	47.1	27.6	32.1	42.2	151
June													
Creel Estimate	0	334.7	23.3	0	3.8	32.0	66.7	56.0	416.6	337.1	74.6	127.6	1,472
Lodge Report	0	415.2	28.8	0	0	0	0	0	0	0	0	0	444
JUNE RETAINED	0	749.9	52.1	0	3.8	32.0	66.7	56.0	416.6	337.1	74.6	127.6	1,916
SE	0	57.1	9.6	0	2.7	9.6	30.0	23.7	51.1	84.5	26.0	37.1	129
July													
Creel Estimate	53.0	1045.2	26.9	0	9.4	116.3	43.4	85.7	712.0	305.8	102.3	26.7	2,527
Lodge Report	30.1	593.6	15.3	0	0	0	0	0	0	0	0	0	639
JULY RETAINED	83.1	1638.9	42.1	0	9.4	116.3	43.4	85.7	712.0	305.8	102.3	26.7	3,166
SE	20.8	114.6	14.4	0	4.2	27.9	12.5	33.3	62.0	64.8	30.6	13.0	158
August													
Creel Estimate	9.6	1053.3	0	0	2.3	102.2	52.1	57.9	326.7	77.7	181.6	41.1	1,905
Lodge Report	2.4	266.6	0.0	0	0	0	0	0	0	0	0	0	269
AUGUST RETAINED	12.0	1319.8	0	0	2.3	102.2	52.1	57.9	326.7	77.7	181.6	41.1	2,174
SE	7.0	170.9	0	0	1.7	35.7	16.0	19.7	47.7	25.6	66.8	16.0	197
TOTAL													
Creel Estimate	191	2,810	60	0	19	324	231	349	1,722	764	437	316	7,223
Lodge Report	72	1,391	47	0	0	0	0	0	0	0	0	0	1,510
TOTAL RETAINED	263	4,201	107	0	19	324	231	349	1,722	764	437	316	8,733
SE	37	228	19	0	6	57	42	100	105	113	84	60	322

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2024.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	40.8	121.3	0	0	7.7	25.6	4.2	0	17.5	0	21.7	0	239
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	40.8	121.3	0	0	7.7	25.6	4.2	0	17.5	0	21.7	0	239
SE	18.2	56.7	0	0	5.6	12.1	3.2	0	8.2	0	12.5	0	63
June													
Creel Estimate	0	23.7	31.0	0	24.9	0	9.1	16.8	52.4	9.1	11.3	0	178
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	23.7	31.0	0	24.9	0	9.1	16.8	52.4	9.1	11.3	0	178
SE	0	8.9	16.1	0	9.6	0	3.7	14.3	16.9	7.6	9.6	0	33
July													
Creel Estimate	0	81.3	0	0	4.7	15.3	10.9	7.1	42.7	15.9	5.0	0	183
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	81.3	0	0	4.7	15.3	10.9	7.1	42.7	15.9	5.0	0	183
SE	0	23.6	0	0	2.4	5.7	3.9	5.6	9.4	9.0	3.5	0	29
August													
Creel Estimate	0	58.1	0	0	4.7	3.4	4.2	0	40.2	33.3	33.6	0	178
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	58.1	0	0	4.7	3.4	4.2	0	40.2	33.3	33.6	0	178
SE	0	32.3	0	0	2.4	2.5	3.1	0	14.0	24.7	20.1	0	48
TOTAL													
Creel Estimate	41	284	31	0	42	44	28	24	153	58	72	0	778
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	41	284	31	0	42	44	28	24	153	58	72	0	778
SE	18	70	16	0	12	14	7	15	25	27	26	0	90

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2024. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	40.8	12.8	20.0	0	19.2	36.6	14.6	0	171.7	51.2	39.1	56.8	463
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	40.8	12.8	20.0	0	19.2	36.6	14.6	0	171.7	51.2	39.1	56.8	463
SE	20.4	6.5	9.0	0	7.4	11.7	6.5	0	51.8	22.1	24.8	29.2	74
June													
Creel Estimate	32.3	3.0	15.5	0	23.0	50.6	30.3	11.2	46.9	227.8	42.1	19.8	503
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	32.3	3.0	15.5	0	23.0	50.6	30.3	11.2	46.9	227.8	42.1	19.8	503
SE	17.2	2.3	8.2	0	8.7	21.1	12.9	9.5	16.5	86.9	15.0	12.1	97
July													
Creel Estimate	145.8	70.2	0	0	23.6	58.2	8.7	0	76.4	19.9	94.8	0	497
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	145.8	70.2	0	0	23.6	58.2	8.7	0	76.4	19.9	94.8	0	497
SE	44.7	17.9	0	0	7.2	17.2	4.3	0	13.6	7.5	22.8	0	59
August													
Creel Estimate	325.7	190.7	111.3	0	16.4	132.9	31.3	29.0	52.1	144.4	242.2	8.6	1,285
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	325.7	190.7	111.3	0	16.4	132.9	31.3	29.0	52.1	144.4	242.2	8.6	1,285
SE	113.9	65.4	78.9	0	6.7	37.3	11.2	20.5	13.5	85.3	64.5	6.1	193
TOTAL													
Creel Estimate	545	277	147	0	82	278	85	40	347	443	418	85	2,747
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	545	277	147	0	82	278	85	40	347	443	418	85	2,747
SE	125	68	80	0	15	48	19	23	58	124	74	32	235

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2024. Included are Big and Longnose skates, Pacific Spiny Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, greenlings, Spotted Ratfish, Arrow-tooth and Starry flounders, Pacific Sanddab, Pacific Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	8.2	19.1	5.0	0	107.7	25.6	18.8	102.2	49.1	21.9	21.7	0	379
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	8.2	19.1	5.0	0	107.7	25.6	18.8	102.2	49.1	21.9	21.7	0	379
SE	4.6	13.8	3.9	0	40.4	8.8	6.0	86.2	17.1	14.3	8.1	0	100
June													
Creel Estimate	12.9	5.9	15.5	0	26.9	45.3	45.5	5.6	91.1	45.6	19.5	8.5	322
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	12.9	5.9	15.5	0	26.9	45.3	45.5	5.6	91.1	45.6	19.5	8.5	322
SE	6.9	3.2	8.2	0	6.1	16.8	17.3	4.8	45.6	25.1	9.3	5.4	97
July													
Creel Estimate	48.6	33.2	0	0	101.3	159.2	54.3	0	60.6	11.9	82.3	0	551
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	48.6	33.2	0	0	101.3	159.2	54.3	0	60.6	11.9	82.3	0	551
SE	25.3	10.1	0	0	20.5	47.7	25.5	0	12.6	6.4	22.1	0	69
August													
Creel Estimate	143.7	124.4	37.1	0.6	51.5	371.4	33.4	0	127.8	55.5	470.8	0	1,416
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	143.7	124.4	37.1	0.6	51.5	371.4	33.4	0	127.8	55.5	470.8	0	1,416
SE	49.1	38.4	26.3	0.4	12.4	109.0	9.4	0	25.4	24.4	199.8	0	241
TOTAL													
Creel Estimate	213	183	58	1	287	602	152	108	329	135	594	9	2,669
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	213	183	58	1	287	602	152	108	329	135	594	9	2,669
SE	56	42	28	0	47	120	33	86	56	38	201	5	276

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by taxon, 2024.

Month / Catch Type	Taxon															TOTAL
	Arrow-tooth Flounder	Big Skate	Cabezon	English Sole	Greenlings	Longnose Skate	Pacific Cod	Pacific Sanddab	Pacific Rock Sole	Spiny Dogfish	Red Irish Lord	Sablefish	Spotted Ratfish	Starry Flounder	Unknown / Other	
May																
Creel Estimate	0	0	11.9	7.0	67.2	3.5	328.6	0	40.7	0	0	0	0	3.8	0	463
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	0	0	11.9	7.0	67.2	3.5	328.6	0	40.7	0	0	0	0	3.8	0	463
SE	0	0	5.4	5.0	19.9	2.5	65.4	0	25.9	0	0	0	0	2.8	0	74
June																
Creel Estimate	0	2.7	1.6	0	31.3	0	429.3	0	16.8	0	0	0	0	5.7	15.2	503
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	2.7	1.6	0	31.3	0	429.3	0	16.8	0	0	0	0	5.7	15.2	503
SE	0	2.2	1.4	0	16.6	0	94.1	0	8.7	0	0	0	0	3.1	10.7	97
July																
Creel Estimate	16.8	3.1	25.5	5.9	26.8	0	387.5	0	3.7	7.5	0	3.1	0	7.1	10.6	497
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	16.8	3.1	25.5	5.9	26.8	0	387.5	0	3.7	7.5	0	3.1	0	7.1	10.6	497
SE	5.1	2.2	11.2	3.1	7.8	0	56.2	0	2.6	3.8	0	2.2	0	3.8	4.5	59
August																
Creel Estimate	2.4	0	0	4.7	49.7	0	1219.7	2.4	0	5.8	0	0	0	0	0	1,285
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	2.4	0	0	4.7	49.7	0	1219.7	2.4	0	5.8	0	0	0	0	0	1,285
SE	1.7	0	0	3.5	15.9	0	191.9	1.7	0	3.0	0	0	0	0	0	193
TOTAL																
Creel Estimate	19	6	39	18	175	4	2,365	2	61	13	0	3	0	17	26	2,747
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	19	6	39	18	175	4	2,365	2	61	13	0	3	0	17	26	2,747
SE	5	3	13	7	31	2	230	2	27	5	0	2	0	6	12	235

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by taxon, 2024.

Month / Catch Type	Taxon															TOTAL
	Arrow-tooth Flounder	Big Skate	Cabezon	English Sole	Greenlings	Longnose Skate	Pacific Cod	Pacific Sanddab	Pacific Rock Sole	Spiny Dogfish	Red Irish Lord	Sablefish	Spotted Ratfish	Starry Flounder	Unknown / Other	
May																
Creel Estimate	28.8	72.8	7.3	0	20.9	3.8	92.7	0	0	111.1	0	3.8	33.8	0	4.2	379
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	28.8	72.8	7.3	0	20.9	3.8	92.7	0	0	111.1	0	3.8	33.8	0	4.2	379
SE	14.7	34.8	3.7	0	8.8	2.8	23.6	0	0	86.8	0	2.8	17.9	0	3.2	100
June																
Creel Estimate	30.3	47.9	12.9	0	21.7	0	72.4	0	20.1	19.4	0	0	38.4	0	59.0	322
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	30.3	47.9	12.9	0	21.7	0	72.4	0	20.1	19.4	0	0	38.4	0	59.0	322
SE	20.4	12.9	8.1	0	13.5	0	26.4	0	12.9	7.9	0	0	13.6	0	40.6	60
July																
Creel Estimate	21.1	66.9	0	0	70.1	2.2	65.9	0	0	272.1	0	0	33.5	4.3	15.3	551
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	21.1	66.9	0	0	70.1	2.2	65.9	0	0	272.1	0	0	33.5	4.3	15.3	551
SE	8.8	12.9	0	0	27.2	1.6	20.5	0	0	56.1	0	0	9.1	3.6	10.9	69
August																
Creel Estimate	53.6	84.4	18.1	5.8	54.6	0	226.6	0	27.3	880.1	0	0	36.4	24.7	4.7	1,416
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	53.6	84.4	18.1	5.8	54.6	0	226.6	0	27.3	880.1	0	0	36.4	24.7	4.7	1,416
SE	12.9	15.7	9.8	3.0	27.1	0	55.9	0	10.8	230.4	0	0	12.3	10.3	3.5	241
TOTAL																
Creel Estimate	134	272	38	6	167	6	458	0	47	1,283	0	4	142	29	83	2,669
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	134	272	38	6	167	6	458	0	47	1,283	0	4	142	29	83	2,669
SE	30	42	13	3	42	3	69	0	17	253	0	3	27	11	42	276

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2024. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	40.8	12.8	20.0	0	19.2	36.6	14.6	0	171.7	51.2	39.1	56.8	463
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RETAINED	40.8	12.8	20.0	0	19.2	36.6	14.6	0	171.7	51.2	39.1	56.8	463
SE	20.4	6.5	9.0	0	7.4	11.7	6.5	0	51.8	22.1	24.8	29.2	74
June													
Creel Estimate	32.3	3.0	15.5	0	23.0	50.6	30.3	11.2	46.9	227.8	42.1	19.8	503
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	32.3	3.0	15.5	0	23.0	50.6	30.3	11.2	46.9	227.8	42.1	19.8	503
SE	17.2	2.3	8.2	0	8.7	21.1	12.9	9.5	16.5	86.9	15.0	12.1	97
July													
Creel Estimate	145.8	70.2	0	0	23.6	58.2	8.7	0	76.4	19.9	94.8	0	497
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	145.8	70.2	0	0	23.6	58.2	8.7	0	76.4	19.9	94.8	0	497
SE	44.7	17.9	0	0	7.2	17.2	4.3	0	13.6	7.5	22.8	0	59
August													
Creel Estimate	325.7	190.7	111.3	0	16.4	132.9	31.3	29.0	52.1	144.4	242.2	8.6	1,285
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	325.7	190.7	111.3	0	16.4	132.9	31.3	29.0	52.1	144.4	242.2	8.6	1,285
SE	113.9	65.4	78.9	0	6.7	37.3	11.2	20.5	13.5	85.3	64.5	6.1	193
TOTAL													
Creel Estimate	545	277	147	0	82	278	85	40	347	443	418	85	2,747
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	545	277	147	0	82	278	85	40	347	443	418	85	2,747
SE	125	68	80	0	15	48	19	23	58	124	74	32	235

Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2024. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
May													
Creel Estimate	63.3	140.4	40.0	1.3	115.4	183.1	156.4	31.5	231.2	87.7	21.7	26.7	1,099
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	63.3	140.4	40.0	1.3	115.4	183.1	156.4	31.5	231.2	87.7	21.7	26.7	1,099
SE	22.1	60.8	31.1	0.7	28.1	47.1	33.7	22.6	45.7	47.9	10.2	18.5	121
June													
Creel Estimate	64.5	234.0	23.3	0.7	149.6	119.9	309.1	78.4	358.7	610.4	13.0	14.2	1,976
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	64.5	234.0	23.3	0.7	149.6	119.9	309.1	78.4	358.7	610.4	13.0	14.2	1,976
SE	48.7	61.3	12.7	0.6	30.6	22.4	59.6	35.3	41.0	213.5	6.1	7.3	245
July													
Creel Estimate	234.2	598.3	62.7	0	171.9	254.1	191.1	32.2	833.3	91.4	84.8	16.4	2,570
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	234.2	598.3	62.7	0	171.9	254.1	191.1	32.2	833.3	91.4	84.8	16.4	2,570
SE	73.3	87.6	25.2	0	44.9	41.1	42.7	17.4	86.1	17.9	22.8	9.0	167
August													
Creel Estimate	124.6	431.3	111.3	0	107.8	231.7	216.9	11.6	359.8	344.2	114.3	15.4	2,069
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	124.6	431.3	111.3	0	107.8	231.7	216.9	11.6	359.8	344.2	114.3	15.4	2,069
SE	44.2	95.5	48.7	0	36.9	40.0	41.9	5.8	52.4	170.4	31.9	5.3	226
TOTAL													
Creel Estimate	486	1,404	237	2	545	789	873	154	1,783	1,134	234	73	7,714
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	486	1,404	237	2	545	789	873	154	1,783	1,134	234	73	7,714
SE	101	156	64	1	71	78	91	46	118	278	41	22	392

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by taxon, 2024.

	Taxon														
Month / Catch Type	Black Rockfish	Boccacio Rockfish	Canary Rockfish	China Rockfish	Copper Rockfish	Dusky Rockfish	Quillback Rockfish	Redband Rockfish	Redstripe Rockfish	Tiger Rockfish	Widow Rockfish	Yelloweye Rockfish	Yellowtail Rockfish	Unknown rockfish	TOTAL
May															
Creel Estimate	377.1	11.2	28.0	55.4	88.0	3.5	300.9	0	0	17.5	2.1	33.7	73.9	298.1	1,289
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	93.0	93
MAY RETAINED	377.1	11.2	28.0	55.4	88.0	3.5	300.9	0	0	17.5	2.1	33.7	73.9	391.1	1,382
SE	74.7	6.0	10.7	13.8	19.8	2.5	41.1	0	0	10.2	1.6	19.3	33.6	68.1	119
June															
Creel Estimate	621.0	5.8	31.6	48.3	69.4	24.7	408.6	0	21.0	18.9	5.8	0	147.9	234.4	1,637
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	110.0	110
JUNE RETAINED	621.0	5.8	31.6	48.3	69.4	24.7	408.6	0	21.0	18.9	5.8	0	147.9	344.4	1,747
SE	153.6	3.0	18.1	18.3	18.6	16.4	61.0	0	15.3	7.1	3.0	0	45.3	74.5	191
July															
Creel Estimate	843.5	11.3	12.9	98.6	87.1	12.4	587.4	0	0	106.1	9.8	2.4	102.9	363.6	2,238
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	227.0	227
JULY RETAINED	843.5	11.3	12.9	98.6	87.1	12.4	587.4	0	0	106.1	9.8	2.4	102.9	590.6	2,465
SE	86.4	4.6	4.3	17.9	15.7	6.3	47.1	0	0	18.7	4.2	1.7	18.1	67.9	125
August															
Creel Estimate	781.6	99.5	2.1	70.6	75.5	37.1	575.7	0	0	79.9	0	16.6	66.8	414.7	2,220
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	91	91
AUGUST RETAINED	781.6	99.5	2.1	70.6	75.5	37.1	575.7	0	0	79.9	0	16.6	66.8	505.7	2,311
SE	113.5	66.5	1.5	20.7	25.9	26.3	67.2	0	0	18.1	0	12.1	17.4	86.4	178
TOTAL															
Creel Estimate	2,623	128	75	273	320	78	1,873	0	21	222	18	53	392	1,311	7,385
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	521	521
TOTAL RETAINED	2,623	128	75	273	320	78	1,873	0	21	222	18	53	392	1,832	7,906
SE	223	67	21	36	41	32	110	0	15	29	5	23	62	149	313

Estimated numbers of rockfish released by lodge and non-lodge anglers, by taxon, 2024.

	Taxon														
Month / Catch Type	Black Rockfish	Boccacio Rockfish	Canary Rockfish	China Rockfish	Copper Rockfish	Dusky Rockfish	Quillback Rockfish	Redband Rockfish	Redstripe Rockfish	Tiger Rockfish	Widow Rockfish	Yelloweye Rockfish	Yellowtail Rockfish	Unknown rockfish	TOTAL
May															
Creel Estimate	62.7	0	3.7	52.0	39.2	0	148.1	10.8	7.0	4.3	4.2	105.8	58.2	602.7	1,099
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MAY RELEASED	62.7	0	3.7	52.0	39.2	0	148.1	10.8	7.0	4.3	4.2	105.8	58.2	602.7	1,099
SE	27.6	0	2.7	25.4	17.3	0	34.9	5.9	5.0	3.1	3.2	36.1	38.9	93.9	121
June															
Creel Estimate	305.5	15.2	13.8	114.1	81.4	16.5	470.8	0	0	16.1	4.7	271.4	21.9	644.6	1,976
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	305.5	15.2	13.8	114.1	81.4	16.5	470.8	0	0	16.1	4.7	271.4	21.9	644.6	1,976
SE	78.0	10.7	6.1	39.8	21.2	6.7	159.8	0	0	6.6	2.4	76.6	8.6	141.5	245
July															
Creel Estimate	175.2	18.4	9.6	200.8	146.7	33.1	390.8	0	0	46.5	0	675.6	34.6	838.9	2,570
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	175.2	18.4	9.6	200.8	146.7	33.1	390.8	0	0	46.5	0	675.6	34.6	838.9	2,570
SE	42.4	9.3	5.5	45.8	38.0	19.0	50.2	0	0	13.8	0	98.7	11.2	97.2	167
August															
Creel Estimate	170.9	0	2.1	39.4	44.0	2.4	376.6	0	0	42.3	0	303.9	49.3	1038.1	2,069
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	170.9	0	2.1	39.4	44.0	2.4	376.6	0	0	42.3	0	303.9	49.3	1038.1	2,069
SE	50.4	0	1.5	11.8	11.3	1.7	87.9	0	0	15.5	0	50.8	19.6	193.1	226
TOTAL															
Creel Estimate	714	34	29	406	311	52	1,386	11	7	109	9	1,357	164	3,124	7,714
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	714	34	29	406	311	52	1,386	11	7	109	9	1,357	164	3,124	7,714
SE	106	14	9	67	48	20	192	6	5	22	4	140	46	275	392

APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month, and subarea, in 2024. Excludes any interviews for which observers were not able to confirm adipose status, or for which biosampling was not a complete sample of fish kept.

Species		Subarea														
Month	Adipose	3I	3J	3K	3?	4A	4B	4C	4D	4E	4F	4G	4H	4L	4?	TOTAL
Chinook Salmon																
May	Clipped	0	0	0	0	0	2	7	2	0	4	0	0	0	0	15
	Not Clipped	0	1	0	0	2	24	32	10	0	17	0	1	0	0	87
	Percent Clipped	-	0%	-	-	0%	8%	18%	17%	-	19%	-	0%	-	-	15%
June	Clipped	0	1	0	0	0	4	6	4	0	4	0	1	0	0	20
	Not Clipped	0	10	0	0	0	16	16	14	0	13	1	0	0	0	70
	Percent Clipped	-	9%	-	-	-	20%	27%	22%	-	24%	0%	100%	-	-	22%
July	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	2	0	0	0	0	0	0	0	1	0	0	0	0	3
	Percent Clipped	-	0%	-	-	-	-	-	-	-	0%	-	-	-	-	0%
August	Clipped	0	0	0	0	0	0	0	1	0	1	0	2	0	0	4
	Not Clipped	0	0	0	0	0	1	5	2	0	3	0	0	0	0	11
	Percent Clipped	-	-	-	-	-	0%	0%	33%	-	25%	-	100%	-	-	27%
Total	Clipped	0	1	0	0	0	6	13	7	0	9	0	3	0	0	39
	Not Clipped	0	13	0	0	2	41	53	26	0	34	1	1	0	0	171
	Percent Clipped	-	7%	-	-	0%	13%	20%	21%	-	21%	0%	75%	-	-	18.6%
Coho Salmon																
May	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Percent Clipped	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
June	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	5	2	0	0	1	25	9	0	16	2	1	0	0	61
	Percent Clipped	-	0%	0%	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%
July	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	5	6	0	0	0	4	14	20	0	56	5	1	0	0	111
	Percent Clipped	0%	0%	-	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%
August	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	0	2	0	0	0	13	16	15	0	8	2	0	0	0	56
	Percent Clipped	-	0%	-	-	-	0%	0%	0%	-	0%	0%	-	-	-	0.0%
Total	Clipped	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Not Clipped	5	13	2	0	0	18	55	44	0	80	9	2	0	0	228
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	-	0%	0%	0%	-	-	0.0%